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# Integrated Services Digital Network (ISDN); Digital Subscriber Signalling System No. one (DSS1) protocol; Network Integration Testing (NIT); ISDN end-to-end testing; Part 1: Test Suite Structure and Test purposes (TSS&TP) specification

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## Foreword

This ETSI Technical Report (ETR) has been produced by the Signalling Protocols and Switching (SPS) Technical Committee of the European Telecommunications Standards Institute (ETSI).

ETRs are informative documents resulting from ETSI studies which are not appropriate for European Telecommunication Standard (ETS) or Interim European Telecommunication Standard (I-ETS) status. An ETR may be used to publish material which is either of an informative nature, relating to the use or the application of ETSs or I-ETSs, or which is immature and not yet suitable for formal adoption as an ETS or an I-ETS.

This ETR is part 1 of a multi-part report covering the Integrated Services Digital Network (ISDN) network integration end-to-end testing, as described below:

**Part 1: "Test Suite Structure and Test Purposes (TSS&TP) specification";**

Part 2: "Implementation Conformance Statement (ICS) proforma, Abstract Test Suite (ATS) and partial Implementation eXtra Information for Testing (IXIT) proforma specification".

## Introduction

This ETR contains the Test Suite Structure and Test Purposes (TSS&TP) developed in the EURESCOM project P412 "Methodology and tools for ISDN Network Integration and Traffic Route Testing" for testing the international European ISDN, covering Network Integration Testing (NIT) between ISDN-ISDN, ISDN-PSTN and PSTN-ISDN networks. The objective is to verify the level of international end-to-end support of ISDN services. Both bearer services (and associated teleservices) and supplementary services are checked for interworking capability and compatibility, in the international European ISDN.

The European ISDN is made up by connecting the different national networks. End-to-end NIT covers all the testing activities necessary to assess the correct behaviour of the interconnected network from the point of view of access interfaces, network side.

Included are the test purposes developed in the preceding EURESCOM project P104 with additional test purposes for the basic call and supplementary services that are supported by the international ISDN User Part (ISUP) versions 1 and 2, and test purposes for end-to-end performance to check B-channel stability and quality. The performance objectives take into consideration the definitions present in the relevant ITU-T Recommendations and adopt the 24 hour measurement period and target values used by ETSI for Open Network Provision (ONP) 64 kbit/s leased line applications.

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# 1 Scope

This first part of ETR 299 specifies the Test Suite Structure and Test Purposes (TSS&TP) for the network side of the T reference point or coincident S and T reference point (as defined in ITU-T Recommendation I.411 [35]) for the pan-European Integrated Services Digital Network (ISDN) as provided by European public telecommunications operators by means of the Digital Subscriber Signalling System No. one (DSS1) protocol, for Network Integration Testing (NIT) covering the end-to-end support of ISDN services.

# 2 References

For the purposes of this ETR, the following references apply:

[1] ETS 300 102-1 (1990) including amendment A2 (1993): "Integrated Services Digital Network (ISDN); User-network interface layer 3; Specifications for basic call control".

[2] ETS 300 083 (1993): "Integrated Services Digital Network (ISDN); Circuit mode structured bearer service category usable for speech information transfer; Terminal requirements necessary for end-to-end compatibility".

[3] ETS 300 084 (1993): "Integrated Services Digital Network (ISDN); Circuit mode structured bearer service category usable for 3,1 kHz audio information transfer; Terminal requirements necessary for end-to-end compatibility".

[4] ETS 300 267-1 (1994) including amendment A1 (1996): "Integrated Services Digital Network (ISDN); Telephony 7 kHz and videotelephony teleservices; Digital Subscriber Signalling System No. one (DSS1) protocol; Part 1: Protocol specification".

[5] ETS 300 080 (1992): "Integrated Services Digital Network (ISDN); ISDN lower layer protocols for telematic terminals".

[6] ETS 300 103 (1990): "Integrated Services Digital Network (ISDN); Support of CCITT Recommendation X.21, X.21 bis and X.20bis based Data Terminal Equipments (DTEs) by an ISDN; Synchronous and asynchronous terminal adaption functions".

[7] ETS 300 092-1 (1992) including amendment A2 (1994): "Integrated Services Digital Network (ISDN); Calling Line Identification Presentation (CLIP) supplementary service; Digital Subscriber Signalling System No. one (DSS1) protocol; Part 1: Protocol specification".

[8] ETS 300 093-1 (1992): "Integrated Services Digital Network (ISDN); Calling Line Identification Restriction (CLIR) supplementary service; Digital Subscriber Signalling System No. one (DSS1) protocol; Part 1: Protocol specification".

[9] ETS 300 097-1 (1992) including amendment A1 (1994): "Integrated Services Digital Network (ISDN); Connected Line Identification Presentation (COLP) supplementary service; Digital Subscriber Signalling System No. one (DSS1) protocol; Part 1: Protocol specification".

[10] ETS 300 098-1 (1992): "Integrated Services Digital Network (ISDN); Connected Line Identification Restriction (COLR) supplementary service; Digital Subscriber Signalling System No. one (DSS1) protocol; Part 1: Protocol specification".

[11] ETS 300 138-1 (1992) including amendment A1 (1996): "Integrated Services Digital Network (ISDN); Closed User Group (CUG) supplementary service; Digital Subscriber Signalling System No. one (DSS1) protocol; Part 1: Protocol specification".

[12] ETS 300 061-1 (1992): "Integrated Services Digital Network (ISDN); Subaddressing (SUB) supplementary service; Digital Subscriber Signalling System No. one (DSS1) protocol; Part 1: Protocol specification".

[13] ETS 300 055-1 (1992): "Integrated Services Digital Network (ISDN); Terminal Portability (TP) supplementary service; Digital Subscriber Signalling System No. one (DSS1) protocol; Part 1: Protocol specification".

[14] ETS 300 286-1 (1996): "Integrated Services Digital Network (ISDN); User-to-User Signalling (UUS) supplementary service; Digital Subscriber Signalling System No. one (DSS1) protocol; Part 1: Protocol specification".

[15] ETS 300 185-1 (1993) including amendment A1 (1995): "Integrated Services Digital Network (ISDN); Conference call, add-on (CONF) supplementary service; Digital Subscriber Signalling System No. one (DSS1) protocol; Part 1: Protocol specification".

[16] ETS 300 207-1 (1994): "Integrated Services Digital Network (ISDN); Diversion supplementary services; Digital Subscriber Signalling System No. one (DSS1) protocol; Part 1: Protocol specification".

[17] ETS 300 210-1 (1996): "Integrated Services Digital Network (ISDN); Freephone (FPH) supplementary service; Digital Subscriber Signalling System No. one (DSS1) protocol; Part 1: Protocol specification".

[18] ETS 300 130-1 (1992): "Integrated Services Digital Network (ISDN); Malicious Call Identification (MCID) supplementary service; Digital Subscriber Signalling System No. one (DSS1) protocol; Part 1: Protocol specification".

[19] ETS 300 188-1 (1993): "Integrated Services Digital Network (ISDN); Three-Party (3PTY) supplementary service; Digital Subscriber Signalling System No. one (DSS1) protocol; Part 1: Protocol specification".

[20] ETS 300 141-1 (1992): "Integrated Services Digital Network (ISDN); Call Hold (HOLD) supplementary service; Digital Subscriber Signalling System No. one (DSS1) protocol; Part 1: Protocol specification".

[21] ETS 300 058-1 (1992): "Integrated Services Digital Network (ISDN); Call Waiting (CW) supplementary service; Digital Subscriber Signalling System No. one (DSS1) protocol; Part 1: Protocol specification".

[22] ETS 300 359-1 (1995): "Integrated Services Digital Network (ISDN); Completion of Calls to Busy Subscriber (CCBS) supplementary service; Digital Subscriber Signalling System No. one (DSS1) protocol; Part 1: Protocol specification".

[23] ETS 300 369-1 (1995): "Integrated Services Digital Network (ISDN); Explicit Call Transfer (ECT) supplementary service; Digital Subscriber Signalling System No. one (DSS1) protocol; Part 1: Protocol specification".

[24] ETS 300 195-1 (1995): "Integrated Services Digital Network (ISDN); Supplementary service interactions; Digital Subscriber Signalling System No. one (DSS1) protocol; Part 1: Protocol specification".

[25] ETS 300 196-1 (1993) including amendment A1 (1995): "Integrated Services Digital Network (ISDN); Generic functional protocol for the support of supplementary services; Digital Subscriber Signalling System No. one (DSS1) protocol; Part 1: Protocol specification".

[26] ETS 300 289 (1994) "Business Telecommunications (BTC); 64 kbit/s digital unrestricted leased line with octet integrity (D64U); Connection characteristics".

[27] TBR 8 (1994): "Integrated Services Digital Network (ISDN); Telephony 3,1 kHz teleservice; Attachment requirements for handset terminals".

[28] ETR 018 (1995): "Integrated Services Digital Network (ISDN); Application of the Bearer Capability (BC), High Layer Compatibility (HLC) and Low Layer Compatibility (LLC) information elements by terminals supporting ISDN services".

[29] ETR 193 (1995): "Methods for Testing and Specification (MTS); Network Integration Testing (NIT); Methodology aspects; Test Co-ordination Procedure (TCP) style guide".

[30] ITU-T Recommendation G.821 (1996): "Error performance of an international digital connection operating at a bit rate below the primary rate and forming part of an Integrated Services Digital Network".

[31] CCITT Recommendation G.822 (1988): "Controlled slip rate objectives of an international digital connection".

[32] CCITT Recommendation O.152 (1988): "Error performance measuring equipment for 64 kbit/s paths".

[33] ITU-T Recommendation I.112 (1993): "Vocabulary and terms for ISDNs".

[34] ITU-T Recommendation I.210 (1993): "Principles of the telecommunication services supported by an ISDN and the means to describe them".

[35] ITU-T Recommendation I.411 (1993): "ISDN user-network interfaces - Reference configurations".

[36] CCITT Recommendation E.164 (1991): "Numbering plan for the ISDN era".

[37] ISO/IEC 9646-1: "Information Technology-OSI Conformance Testing Methodology and Framework, Part 1: General Concepts".

[38] ISO/IEC 9646-2: "Information Technology-OSI Conformance Testing Methodology and Framework, Part 2: Abstract Test Suite Specification".

[39] ISO/IEC 9646-3: "Information Technology-OSI Conformance Testing Methodology and Framework, Part 3: The Tree and Tabular Combined Notation".

# 3 Definitions

For the purposes of this ETR, the following definitions apply:

## 3.1 Definitions related to conformance testing

**abstract test case:** Refer to ISO/IEC 9646-1 [37].

**abstract test suite:** Refer to ISO/IEC 9646-1 [37].

**Implementation Under Test (IUT):** Refer to ISO/IEC 9646-1 [37].

**lower tester:** Refer to ISO/IEC 9646-1 [37].

**Protocol Implementation Conformance Statement (PICS):** Refer to ISO/IEC 9646-1 [37].

**PICS proforma:** Refer to ISO/IEC 9646-1 [37].

**Protocol Implementation eXtra Information for Testing (PIXIT):** Refer to ISO/IEC 9646-1 [37].

**PIXIT proforma:** Refer to ISO/IEC 9646-1 [37].

**point of control and observation:** Refer to ISO/IEC 9646-1 [37].

**system under test:** Refer to ISO/IEC 9646-1 [37].

**test purpose:** Refer to ISO/IEC 9646-1 [37].

### 3.2 Definitions related to ETS 300 102-1

**Integrated Services Digital Network (ISDN):** See ITU-T Recommendation I.112 [33], definition 308.

**ISDN number:** A number conforming to the numbering and structure specified in CCITT Recommendation E.164 [36].

**service; telecommunication service:** See ITU-T Recommendation I.112 [33], definition 201.

**supplementary service:** See ITU-T Recommendation I.210 [34], subclause 2.4.

**user:** The DSS1 protocol entity at the User side of the user-network interface where a T reference point or coincident S and T reference point applies.

**user (S/T):** The DSS1 protocol entity at the User side of the user-network interface where a coincident S and T reference point applies.

**user (T):** The DSS1 protocol entity at the User side of the user-network interface where a T reference point applies (User is the Private ISDN).

### 3.3 Definitions related to test purpose descriptions

**BC = 3,1 kHz audio:** A Bearer capability information element with its information transfer capability field set to "3,1 kHz Audio" and its user information layer one protocol field set to "G.711 A-law" (see ETS 300 084 [3]).

**BC = speech:** A Bearer capability information element with its information transfer capability field set to "speech" and its user information layer one protocol field set to "G.711 A-law" (see ETS 300 083 [2]).

**BC = UDI/TA:** A Bearer capability information element with its information transfer capability set to "unrestricted digital information with tones/announcements" and its user information layer one protocol field set to "Recommendations H.221 and H.242" (see ETS 300 267-1 [4]).

**BC = UDI:** A Bearer capability information element with its information transfer capability set to "unrestricted digital information" (see ETS 300 102-1 [1]).

**BC = V110/X30:** A Bearer capability information element with its information transfer capability set to "unrestricted digital information" and its user information layer 1 field set to "CCITT standardized rate adaption V.110/X.30", including sync./async. and user rate values (see ETS 300 102-1 [1]).

**CF active:** The call forwarding (U, B or NR) supplementary service is already activated with the address of user C (see ETS 300 207-1 [16]).

**CUG default request:** The calling user does not include an explicit request for the CUG supplementary service in the outgoing SETUP message (see ETS 300 138-1 [11]).

**errored second:** A second with one or more bit errors (see ITU-T Recommendation G.821 [30]).

**Errored Seconds Ratio (ESR):** The ratio of errored seconds over all seconds within a specified measuring period, where neither are counted during unavailability periods (see ITU-T Recommendation G.821 [30]).

**HLC = facsimile group 2/3:** A High Layer compatibility information element with its high layer characteristics identification field set to "facsimile group 2/3 (Rec. F.182)" (see ETS 300 102-1 [1]).

**HLC = facsimile group 4:** A High Layer compatibility information element with its high layer characteristics identification field set to "facsimile group 4 class 1" (see ETS 300 102-1 [1] and ETS 300 080 [5]).

**HLC = telephony:** A High Layer compatibility information element with its high layer characteristics identification field set to "telephony" (see TBR 008 [27]).

**HLC = telex:** A High Layer compatibility information element with its high layer characteristics identification field set to "telex" (see ETS 300 102-1 [1]).

**HLC = videotelephony_ic:** A High Layer compatibility information element with its high layer characteristics identification field set to "videotelephony (Rec. F.721)" and its extended audiovisual characteristics field set to "capability set of initial channel of Rec. H.221" (see ETS 300 267-1 [4]).

**LLC = telematic_term:** A Low Layer compatibility information element with its user information layer 2 field indicating "ISO/IEC 7776 DTE-DTE operation" and user information layer 3 field indicating "ISO/IEC 8208" (see ETS 300 102-1 [1] and ETS 300 080 [5]).

**LLC = V110/X30:** A Low Layer compatibility information element with its user information layer 1 field indicating "CCITT standardized rate adaption V.110/X.30" and including sync./async. and user rate values (see ETS 300 103 [6]).

**LLC = voice band data via modem:** A Low Layer compatibility information element with its user information layer 1 field indicating a "modem type" coding (see ETS 300 102-1 [1]).

**NPI = unknown:** Numbering plan identification forwarded to the served user coded as "unknown" (see ETS 300 102-1 [1]).

**octet slip:** A slip of one complete octet (see CCITT Recommendation G.822 [31]).

**PI = PR:** Presentation Indicator forwarded to the served user coded as "Presentation restricted" (see ETS 300 102-1 [1]).

**Pseudo-Random Binary Sequence** (**PRBS) = $2^{11}$-1:** Pseudo-random binary sequence according to CCITT Recommendation O.152 [32] transmitted for two consecutive periods of 24 hours. If an unavailability period of more than one hour occurred during the measuring period, it shall be extended accordingly.

**severely errored second:** A second where at least 0,1 % of the bits are errored (corresponds to a one-second interval with a bit-error ratio worse than $1 \times 10^{-3}$) (see ITU-T Recommendation G.821 [30]).

**Severely Errored Seconds Ratio (SESR):** The ratio of severely errored seconds over all seconds within a specified measuring period, where neither are counted during unavailability periods (see ITU-T Recommendation G.821 [30]).

**SI = NP:** Screening Indicator forwarded to the served user coded as "Network provided" (see ETS 300 102-1 [1]).

**SI = UPVP:** Screening Indicator forwarded to the served user coded as "User-provided, verified and passed" (see ETS 300 102-1 [1]).

**slip:** One or more extra or missing consecutive unit intervals in the bit stream (see CCITT Recommendation G.822 [31]).

**telephony 7 kHz fallback not allowed SETUP message:** A SETUP message containing a single BC = UDI/TA and a HLC = telephony (see ETS 300 267-1 [4]).

**TON = international:** Type of number forwarded to the served user coded as "international" (see ETS 300 102-1 [1]).

**TON = unknown:** Type of number forwarded to the served user coded as "unknown" (see ETS 300 102-1 [1]).

**UI length = 32:** The length of the User information field of the User-user information element is 32 octets (see ETS 300 286-1 [14]).

**unavailability period:** A period of time beginning at the first of 10 consecutive severely errored seconds and ending immediately before the first following period of 10 consecutive seconds none of which are severely errored (see CCITT Recommendation G.822 [31]).

**videotelephony fallback not allowed SETUP message:** A SETUP message containing a single BC = UDI/TA and a single HLC = videotelephony_ic (see ETS 300 267-1 [4]).

## 4 Abbreviations

For the purposes of this ETR, the following abbreviations apply:

| | |
|---|---|
| 3PTY | Three-party conference |
| BC | Bearer Capability information element |
| CCBS | Completion of Calls to Busy Subscriber |
| CD | Call Deflection |
| CFB | Call Forwarding Busy |
| CFNR | Call Forwarding No Reply |
| CFU | Call Forwarding Unconditional |
| CLIP | Calling Line Identification Presentation |
| CLIR | Calling Line Identification Restriction |
| COLP | Connected Line identification Presentation |
| COLR | Connected Line identification Restriction |
| CONF | Conference call, add-on |
| CUG | Closed User Group |
| CW | Call Waiting |
| DSS1 | Digital Subscriber Signalling System No. one |
| ECT | Explicit Call Transfer |
| ESR | Errored Seconds Ratio |
| FPH | Freephone |
| FTAM | File Transfer Access and Management |
| HLC | High Layer Compatibility information element |
| ISDN | Integrated Services Digital Network |
| IUT | Implementation Under Test |
| LLC | Low Layer Compatibility information element |
| MCID | Malicious Call Identification |
| NIT | Network Integration Testing |
| ONP | Open Network Provision |
| OSI | Open Systems Interconnection |
| P104 | EURESCOM P104 ATS v1.2 |
| PI | Presentation Indicator |
| PICS | Protocol Implementation Conformance Statement |
| PIXIT | Protocol Implementation eXtra Information for Testing |
| PRBS | Pseudo-Random Binary Sequence |
| SESR | Severely Errored Seconds Ratio |
| SI | Screening Indicator |
| SUB | Subaddressing |
| TC | Test Case |
| TON | Type Of Number |
| TP | Terminal Portability |
| TSS&TP | Test Suite Structure and Test Purposes |
| UDI | Unrestricted Digital Information |
| UDI-TA | UDI with Tones/Announcements |
| UUS | User-to-User Signalling |
| UUS1 | UUS service 1 |
| UUS2 | UUS service 2 |
| UUS3 | UUS service 3 |

# 5 Test suite structure

## 5.1 ISDN-ISDN

| | | | | |
|---|---|---|---|---|
| **ISDN-ISDN** | **Basic_call** (1) | **Successful** (1) | Speech | 1101xx |
| | | | UDI | 1102xx |
| | | | Audio | 1103xx |
| | | | UDI-TA | 1104xx |
| | | **Unsuccessful** (2) | Speech | 1201xx |
| | | | UDI | 1202xx |
| | | | Audio | 1203xx |
| | | | UDI-TA | 1204xx |
| | **Supplementary_ services** (2) | **Speech** (1) | CLIP | 2101xx |
| | | | CLIR | 2102xx |
| | | | COLP | 2103xx |
| | | | COLR | 2104xx |
| | | | CUG | 2105xx |
| | | | SUB | 2106xx |
| | | | TP | 2107xx |
| | | | UUS | 2108xx |
| | | | CONF | 2109xx |
| | | | CFU | 2111xx |
| | | | CFB | 2112xx |
| | | | CFNR | 2113xx |
| | | | CD | 2114xx |
| | | | FPH | 2115xx |
| | | | MCID | 2116xx |
| | | | 3PTY | 2117xx |
| | | | HOLD | 2118xx |
| | | | CW | 2119xx |
| | | | ECT | 2121xx |
| | | | Comb | 2130xx |
| | | **UDI** (2) | CLIP | 2201xx |
| | | | CLIR | 2202xx |
| | | | COLP | 2203xx |
| | | | COLR | 2204xx |
| | | | CUG | 2205xx |
| | | | SUB | 2206xx |
| | | | TP | 2207xx |
| | | | UUS | 2208xx |
| | | | CFU | 2211xx |
| | | | CFB | 2212xx |
| | | | CFNR | 2213xx |
| | | | CD | 2214xx |
| | | | FPH | 2215xx |
| | | | MCID | 2216xx |
| | | | HOLD | 2218xx |
| | | | CW | 2219xx |
| | | | ECT | 2221xx |
| | | | Comb | 2230xx |

**Figure 1 (continued): Test suite structure for ISDN-ISDN**

| | | | | |
|---|---|---|---|---|
| **B-channel** (3) | (0) | | Speech | 3001xx |
| | | | UDI | 3002xx |
| | | | Audio | 3003xx |
| | | | UDI-TA | 3004xx |

NOTE: Numbers in brackets represent group/subgroup numbers and are used in test purpose identifiers.

**Figure 1 (concluded): Test suite structure for ISDN-ISDN**

## 5.2 ISDN-PSTN

| | | | | |
|---|---|---|---|---|
| **ISDN-PSTN** | **Basic_call** (4) | **Successful** (1) | Speech | 4101xx |
| | | | Audio | 4103xx |
| | | **Unsuccessful** (2) | Speech | 4201xx |
| | | | UDI | 4202xx |
| | | | Audio | 4203xx |
| | **Supplementary_ services** (5) | **Speech** (1) | | 5100xx |

NOTE: Numbers in brackets represent group/subgroup numbers and are used in test purpose identifiers.

**Figure 2: Test suite structure for ISDN-PSTN**

## 5.3 PSTN-ISDN

| | | | |
|---|---|---|---|
| **PSTN-ISDN** | **Basic_call** (6) | **Successful** (1) | 6101xx |
| | | **Unsuccessful** (2) | 6201xx |
| | **Supplementary_ services** (7) | **Speech** (1) | 7100xx |

NOTE: Numbers in brackets represent group/subgroup numbers and are used in test purpose identifiers.

**Figure 3: Test suite structure for PSTN-ISDN**

# 6 Test purposes

## 6.1 Introduction

For each test requirement a test purpose is defined.

### 6.1.1 Test purpose naming convention

The test purposes are identified by a six figure numbering scheme where the first figure identifies the test group, followed by a three figure number for subgroup and a two figure serial number (starting at 01 within each group/subgroup). Groups are organized according to the TSS (see table 1).

**Table 1: Test purpose identifier naming convention scheme**

| | |
|---|---|
| Identifier: | **TC <test group > <subgroup> <nn>** |
| <test group>= | 1 digit field representing group reference according to TSS<br>1 = ISDN-ISDN/Basic_call<br>2 = ISDN-ISDN/Supplementary_services<br>3 = ISDN-ISDN/B-channel<br>4 = ISDN-PSTN/Basic_call<br>5 = ISDN-PSTN/Supplementary_services<br>6 = PSTN-ISDN/Basic_call<br>7 = PSTN-ISDN/Supplementary_services |
| <subgroup> = | 3 digit field representing sub group reference according to TSS |
| <nn> = | sequential number (01-99) |

### 6.1.2 Source of test purpose definition

The test purposes are based on ETS 300 102-1 [1] and the applicable standards for ISDN supplementary services ([7] to [25]).

### 6.1.3 Test purpose structure

The test purposes are formatted as tables using the format shown in table 2. Text in bold shows the text which is always present. Plain text provides a description of each field.

**Table 2: Format of a single test purpose**

| **Identifier** | **Ref. to ETS 300 102-1 [1]** | **Other relevant ref.:** |
|---|---|---|
| **TSS reference:** | Test suite structure reference | |
| **Selection criteria:** | The criteria necessary in order to select the test | |
| **Test purpose:** | Description of the test purpose | |
| **Parameter values:** | Values of parameters used for the test execution. For an explanation of the abbreviations used, see subclause 3.3 | |
| **Comments:** | Any relevant comments and reference to equivalent P104 TC if applicable | |

### 6.1.4 Test strategy

As the base standards ([1] and [7] to [25]) contain no explicit requirements for testing, the test purposes were generated as a result of an analysis of the base standards and the PICS specifications. The criteria applied include the following:

- only the requirements from the point of view of the T or coincident S and T reference point are considered;

- whether or not a test case can be built from the test purpose is not considered.

### 6.1.5 End-to-end performance objectives

ITU-T Recommendation G.821 [30] defines performance objectives to be meet for each direction of a N × 64 kbit/s ($1 \leq N < 32$) circuit-switched connection used for voice or data-type traffic. The G.821 error performance objectives for international ISDN connections are presented in table 3. A measuring period of one month is suggested as a reference.

**Table 3: G.821 error performance objectives**

| Performance parameters | Objective |
|---|---|
| Severely errored seconds ratio | Fewer than 0,2 % of one-second intervals to have a bit-error ratio worse than $1 \times 10^{-3}$ |
| Errored seconds ratio | Fewer than 8 % of one-second intervals to have any errors |

ETS 300 289 [26] specifies the technical requirements (and test principles) for the connection characteristics of Open Network Provision (ONP) 64 kbit/s digital unrestricted leased lines with octet integrity. The end-to-end performance objectives are derived from ITU-T Recommendation G.821 [30]: the test values have been transformed to fit a measuring period of 24 hours instead of one month, taking also in consideration the circuit configurations relevant for Europe (see ETS 300 289 [26], annex B).

For the ISDN bearer services "unrestricted digital information" and "unrestricted digital information with tones/announcements", the connection characteristics of the B-channel established between users may be considered equivalent in service to a ONP 64 kbit/s digital unrestricted leased line, because the ISDN circuit connection provides equivalent access to the full digital bit rate of 64 kbit/s with no restrictions on the binary content, using network timing for both directions of the transmission. For those bearer services, it is proposed to adopt the performance objectives defined in ETS 300 289 [26] for the performance of the end-to-end B-channel connection. Table 4 presents these objectives, using a 24 hour measurement period.

**Table 4: B-channel performance objectives**

| Error parameters | Value per 24 hour period |
|---|---|
| Octet slip | < 5 |
| Errored seconds | < 5 324 |
| Severely errored seconds | < 105 |

**Octet slip**

Requirement: For at least one of two consecutive 24 hour measuring periods, the number of octet slips shall be less than 5 (slips other than octet slips are considered as errors).

NOTE 1: This requirement is based on CCITT Recommendation G.822 [31], clause 2 and table 1/G.822.

**Errored seconds**

Requirement: For at least one of two consecutive 24 hour measuring periods, the number of errored seconds shall be less than 5 324.

NOTE 2: This 24 hour test limit corresponds to a mean errored seconds ratio of $6,4 \times 10^{-2}$.

**Severely errored seconds**

Requirement: For at least one of two consecutive 24 hour measuring periods, the number of severely errored seconds shall be less than 105.

NOTE 3: This 24 hour test limit corresponds to a mean severely errored seconds ratio of $1,5 \times 10^{-3}$.

## 6.2 Test purposes

### 6.2.1 Test purposes for ISDN-ISDN

#### 6.2.1.1 Test purposes for ISDN-ISDN, basic call

| 110101 | Ref. to ETS 300 102-1 [1], subclause 5.1.5.1 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Successful/Speech/TC110101 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that call establishment using en-bloc sending is performed correctly | |
| **Parameter values:** | BC=speech, no HLC | |
| **Comments:** | P104 TC11101 | |

| 110102 | Ref. to ETS 300 102-1 [1], subclause 5.1.5.2 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Successful/Speech/TC110102 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that call establishment using overlap sending is performed correctly | |
| **Parameter values:** | BC=speech, no HLC | |
| **Comments:** | P104 TC11102 | |

| 110103 | Ref. to ETS 300 102-1 [1], subclause 5.3.3 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Successful/Speech/TC110103 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that the call clearing procedure is performed correctly when the calling user clears after answer | |
| **Parameter values:** | BC=speech, no HLC | |
| **Comments:** | P104 TC11103 | |

| 110104 | Ref. to ETS 300 102-1 [1], subclause 5.3.3 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Successful/Speech/TC110104 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that the call clearing procedure is performed correctly when the called user clears after answer | |
| **Parameter values:** | BC=speech, no HLC | |
| **Comments:** | P104 TC11104 | |

| 110105 | Ref. to ETS 300 102-1 [1], subclause 4.5.16 | Other relevant ref.: TBR 8 [27], subclause 5.1.3, ETR 018 [28], subclause 6.3.1 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Successful/Speech/TC110105 | |
| **Selection criteria:** | Telephony 3,1 kHz teleservice | |
| **Test purpose:** | Support of telephony 3,1 kHz teleservice: Ensure that the HLC information is transported transparently through the network and correctly delivered to the called user | |
| **Parameter values:** | BC=speech, HLC = telephony | |
| **Comments:** | P104 TC11105 | |

| 110201 | Ref. to ETS 300 102-1 [1], subclause 5.1.5.1 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Successful/UDI/TC110201 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that call establishment using en-bloc sending is performed correctly | |
| **Parameter values:** | BC=UDI, no HLC | |
| **Comments:** | P104 TC11301 | |

| 110202 | Ref. to ETS 300 102-1 [1], subclause 5.1.5.2 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Successful/UDI/TC110202 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that call establishment using overlap sending is performed correctly | |
| **Parameter values:** | BC=UDI, no HLC | |
| **Comments:** | P104 TC11302 | |

| 110203 | Ref. to ETS 300 102-1 [1], subclause 5.3.3 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Successful/UDI/TC110203 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that the call clearing procedure is performed correctly when the calling user clears after answer | |
| **Parameter values:** | BC=UDI, no HLC | |
| **Comments:** | P104 TC11303 | |

| 110204 | Ref. to ETS 300 102-1 [1], subclause 5.3.3 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Successful/UDI/TC110204 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that the call clearing procedure is performed correctly when the called user clears after answer | |
| **Parameter values:** | BC=UDI, no HLC | |
| **Comments:** | P104 TC11304 | |

| 110205 | Ref. to ETS 300 102-1 [1], subclause 4.5.16 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Successful/UDI/TC110205 | |
| **Selection criteria:** | Telefax G4 teleservice | |
| **Test purpose:** | Support of Telefax G4 teleservice (no LLC): Ensure that the HLC information is transported transparently through the network and correctly delivered to the called user | |
| **Parameter values:** | BC=UDI, HLC=facsimile group 4, no LLC | |
| **Comments:** | P104 TC11305 | |

| 110206 | Ref. to ETS 300 102-1 [1], subclauses 4.5.16, 4.5.18 | Other relevant ref.: ETS 300 080 [5], subclause 4.5.2.1, ETR 018 [28], subclause 6.3.2 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Successful/UDI/TC110206 | |
| **Selection criteria:** | Telefax G4 teleservice | |
| **Test purpose:** | Support of telefax G4 teleservice: Ensure that the LLC and HLC information is transported transparently through the network and correctly delivered to the called user | |
| **Parameter values:** | BC=UDI, HLC=facsimile group 4, LLC=telematic_term | |
| **Comments:** | P104 TC11306 | |

| 110207 | Ref. to ETS 300 102-1 [1], subclause 4.5.16 | Other relevant ref.: ETR 018 [28], subclause 7.1.3 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Successful/UDI/TC110207 | |
| **Selection criteria:** | Teletex terminal (basic and mixed mode) | |
| **Test purpose:** | Support of teletex basic and mixed mode terminals: Ensure that the LLC and HLC information is transported transparently through the network and correctly delivered to the called user | |
| **Parameter values:** | BC=UDI, HLC = teletex mixed mode, LLC=telematic_term | |
| **Comments:** | P104 TC11307 | |

| 110208 | Ref. to ETS 300 102-1 [1], subclause 4.5.16 | Other relevant ref.: ETR 018 [28], subclause 7.1.3 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Successful/UDI/TC110208 | |
| **Selection criteria:** | Teletex terminal (basic and processable mode) | |
| **Test purpose:** | Support of teletex basic and processable mode terminals: Ensure that the LLC and HLC information is transported transparently through the network and correctly delivered to the called user | |
| **Parameter values:** | BC=UDI, HLC = teletex processable, LLC=telematic_term | |
| **Comments:** | P104 TC11308 | |

| 110209 | Ref. to ETS 300 102-1 [1], subclause 4.5.16 | Other relevant ref.: ETR 018 [28], subclause 7.1.3 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Successful/UDI/TC110209 | |
| **Selection criteria:** | Teletex terminal (basic mode) | |
| **Test purpose:** | Support of teletex basic mode terminals: Ensure that the LLC and HLC information is transported transparently through the network and correctly delivered to the called user | |
| **Parameter values:** | BC=UDI, HLC = teletex basic, LLC=telematic_term | |
| **Comments:** | P104 TC11309 | |

| 110210 | Ref. to ETS 300 102-1 [1], subclause 4.5.16 | Other relevant ref.: ETS 300 080 [5], subclause 4.5.2.1 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Successful/UDI/TC110210 | |
| **Selection criteria:** | International videotex interworking | |
| **Test purpose:** | Support of international videotex interworking: Ensure that the LLC and HLC information is transported transparently through the network and correctly delivered to the called user | |
| **Parameter values:** | BC=UDI, HLC=videotex interworking, LLC=telematic_term | |
| **Comments:** | P104 TC11310 | |

| 110211 | Ref. to ETS 300 102-1 [1], subclause 4.5.16 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Successful/UDI/TC110211 | |
| **Selection criteria:** | Telex service | |
| **Test purpose:** | Support of telex service: Ensure that the HLC information is transported transparently through the network and correctly delivered to the called user | |
| **Parameter values:** | BC=UDI, HLC=telex, no LLC | |
| **Comments:** | P104 TC11311 | |

| 110212 | Ref. to ETS 300 102-1 [1], subclause 4.5.16 | Other relevant ref.: ETS 300 080 [5], subclause 4.5.2.1 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Successful/UDI/TC110212 | |
| **Selection criteria:** | Message Handling Systems | |
| **Test purpose:** | Support of Message Handling Systems: Ensure that the HLC information is transported transparently through the network and correctly delivered to the called user | |
| **Parameter values:** | BC=UDI, HLC=message handling system, no LLC | |
| **Comments:** | P104 TC11312 | |

| 110213 | Ref. to ETS 300 102-1 [1], subclause 4.5.16 | Other relevant ref.: ETS 300 080 [5], subclause 4.5.2.1 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Successful/UDI/TC110213 | |
| **Selection criteria:** | OSI applications | |
| **Test purpose:** | Support of OSI application: Ensure that the HLC information is transported transparently through the network and correctly delivered to the called user | |
| **Parameter values:** | BC=UDI, HLC=OSI application, no LLC | |
| **Comments:** | P104 TC11313 | |

| 110214 | Ref. to ETS 300 102-1 [1], subclause 4.5.16 | Other relevant ref.: ETS 300 267-1 [4], clause 7 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Successful/UDI/TC110214 | |
| **Selection criteria:** | Videotelephony teleservice | |
| **Test purpose:** | Support of videotelephony teleservice: Ensure that the HLC information is transported transparently through the network and correctly delivered to the called user (Note). | |
| **Parameter values:** | BC=UDI, HLC=videotelephony_ic | |
| **Comments:** | P104 TC11314<br>Note: support of videotelephony using circuit-mode 64 kbit/s unrestricted digital information. This special situation is considered in ETS 300 267-1 [4], clause 7, note 1. | |

| 110215 | Ref. to ETS 300 102-1 [1], subclause 4.5.5 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Successful/UDI/TC110215 | |
| **Selection criteria:** | V.110/X.30 rate adaption | |
| **Test purpose:** | Support of terminal adaptors V.110/X.30: Ensure that the BC information is transported transparently through the network and correctly delivered to the called user. | |
| **Parameter values:** | BC=V110/X30 synchronous user rate 2,4 kbit/s, no LLC | |
| **Comments:** | P104 TC11315 | |

| 110216 | Ref. to ETS 300 102-1 [1], subclause 4.5.5 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Successful/UDI/TC110216 | |
| **Selection criteria:** | V.110/X.30 rate adaption | |
| **Test purpose:** | Support of terminal adaptors V.110/X.30: Ensure that the BC information is transported transparently through the network and correctly delivered to the called user. | |
| **Parameter values:** | BC=V110/X30 synchronous user rate 9,6 kbit/s, no LLC | |
| **Comments:** | P104 TC11316 | |

| 110217 | Ref. to ETS 300 102-1 [1], subclause 4.5.5 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Successful/UDI/TC110217 | |
| **Selection criteria:** | V.110/X.30 rate adaption | |
| **Test purpose:** | Support of terminal adaptors V.110/X.30: Ensure that the BC information is transported transparently through the network and correctly delivered to the called user. | |
| **Parameter values:** | BC=V110/X30 synchronous user rate 19,2 kbit/s, no LLC | |
| **Comments:** | P104 TC11317 | |

| 110218 | Ref. to ETS 300 102-1 [1], subclause 4.5.5, 4.5.18 | Other relevant ref.: ETS 300 103 [6], annex I, ETR 018 [28], subclause 7.1.1 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Successful/UDI/TC110218 | |
| **Selection criteria:** | V.110/X.30 rate adaption | |
| **Test purpose:** | Support of terminal adaptors V.110/X.30: Ensure that the LLC information is transported transparently through the network and correctly delivered to the called user. | |
| **Parameter values:** | BC=UDI, LLC=V110/X30 synchronous user rate 2,4 kbit/s | |
| **Comments:** | P104 TC11318 | |

| 110219 | Ref. to ETS 300 102-1 [1], subclause 4.5.18 | Other relevant ref.: ETS 300 103 [6], annex I, ETR 018 [28], subclause 7.1.1 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Successful/UDI/TC110219 | |
| **Selection criteria:** | V.110/X.30 rate adaption | |
| **Test purpose:** | Support of terminal adaptors V.110/X.30: Ensure that the LLC information is transported transparently through the network and correctly delivered to the called user. | |
| **Parameter values:** | BC=UDI, LLC=V110/X30 synchronous user rate 9,6 kbit/s | |
| **Comments:** | P104 TC11319 | |

| 110220 | Ref. to ETS 300 102-1 [1], subclause 4.5.18 | Other relevant ref.: ETS 300 103 [6], annex I, ETR 018 [28], subclause 7.1.1 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Successful/UDI/TC110220 | |
| **Selection criteria:** | V.110/X.30 rate adaption | |
| **Test purpose:** | Support of terminal adaptors V.110/X.30: Ensure that the LLC information is transported transparently through the network and correctly delivered to the called user. | |
| **Parameter values:** | BC=UDI, LLC=V110/X30 synchronous user rate 19,2 kbit/s | |
| **Comments:** | P104 TC11320 | |

| 110221 | Ref. to ETS 300 102-1 [1], subclause 4.5.5, 4.5.18 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Successful/UDI/TC110221 | |
| **Selection criteria:** | V.110/X.30 rate adaption | |
| **Test purpose:** | Support of terminal adaptors V.110/X.30: Ensure that BC and LLC information is transported transparently through the network and correctly delivered to the called user. | |
| **Parameter values:** | BC=LLC=V110/X30 synchronous user rate 2,4 kbit/s | |
| **Comments:** | | |

| 110222 | Ref. to ETS 300 102-1 [1], subclause 4.5.5, 4.5.18 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Successful/UDI/TC110222 | |
| **Selection criteria:** | V.110/X.30 rate adaption | |
| **Test purpose:** | Support of terminal adaptors V.110/X.30: Ensure that BC and LLC information is transported transparently through the network and correctly delivered to the called user. | |
| **Parameter values:** | BC=LLC=V110/X30 synchronous user rate 9,6 kbit/s | |
| **Comments:** | | |

| 110223 | Ref. to ETS 300 102-1 [1], subclauses 4.5.5, 4.5.18 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Successful/UDI/TC110223 | |
| **Selection criteria:** | V.110/X.30 rate adaption | |
| **Test purpose:** | Support of terminal adaptors V.110/X.30: Ensure that BC and LLC information is transported transparently through the network and correctly delivered to the called user. | |
| **Parameter values:** | BC=LLC=V110/X30 synchronous user rate 19,2 kbit/s | |
| **Comments:** | | |

| 110224 | Ref. to ETS 300 102-1 [1], subclause 4.5.5 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Successful/UDI/TC110224 | |
| **Selection criteria:** | V.110/X.30 rate adaption | |
| **Test purpose:** | Support of terminal adaptors V.110/X.30: Ensure that the BC information is transported transparently through the network and correctly delivered to the called user. | |
| **Parameter values:** | BC=V110/X30 asynchronous user rate 2,4 kbit/s, no LLC | |
| **Comments:** | P104 TC11321 | |

| 110225 | Ref. to ETS 300 102-1 [1], subclause 4.5.5 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Successful/UDI/TC110225 | |
| **Selection criteria:** | V.110/X.30 rate adaption | |
| **Test purpose:** | Support of terminal adaptors V.110/X.30: Ensure that the BC information is transported transparently through the network and correctly delivered to the called user. | |
| **Parameter values:** | BC=V110/X30 asynchronous user rate 9,6 kbit/s, no LLC | |
| **Comments:** | P104 TC11322 | |

| 110226 | Ref. to ETS 300 102-1 [1], subclause 4.5.5 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Successful/UDI/TC110226 | |
| **Selection criteria:** | V.110/X.30 rate adaption | |
| **Test purpose:** | Support of terminal adaptors V.110/X.30: Ensure that the BC information is transported transparently through the network and correctly delivered to the called user. | |
| **Parameter values:** | BC=V110/X30 asynchronous user rate 19,2 kbit/s, no LLC | |
| **Comments:** | P104 TC11323 | |

| 110227 | Ref. to ETS 300 102-1 [1], subclause 4.5.18 | Other relevant ref.: ETS 300 103 [6], annex I, ETR 018 [28], subclause 7.1.1 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Successful/UDI/TC110227 | |
| **Selection criteria:** | V.110/X.30 rate adaption | |
| **Test purpose:** | Support of terminal adaptors V.110/X.30: Ensure that the LLC information is transported transparently through the network and correctly delivered to the called user. | |
| **Parameter values:** | BC=UDI, LLC=V110/X30 asynchronous user rate 2,4 kbit/s | |
| **Comments:** | P104 TC11324 | |

| 110228 | Ref. to ETS 300 102-1 [1], subclause 4.5.18 | Other relevant ref.: ETS 300 103 [6], annex I, ETR 018 [28], subclause 7.1.1 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Successful/UDI/TC110228 | |
| **Selection criteria:** | V.110/X.30 rate adaption | |
| **Test purpose:** | Support of terminal adaptors V.110/X.30: Ensure that the LLC information is transported transparently through the network and correctly delivered to the called user. | |
| **Parameter values:** | BC=UDI, LLC=V110/X30 asynchronous user rate 9,6 kbit/s | |
| **Comments:** | P104 TC11325 | |

| 110229 | Ref. to ETS 300 102-1 [1], subclause 4.5.18 | Other relevant ref.: ETS 300 103 [6], annex I, ETR 018 [28], subclause 7.1.1 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Successful/UDI/TC110229 | |
| **Selection criteria:** | V.110/X.30 rate adaption | |
| **Test purpose:** | Support of terminal adaptors V.110/X.30: Ensure that the LLC information is transported transparently through the network and correctly delivered to the called user. | |
| **Parameter values:** | BC=UDI, LLC=V110/X30 asynchronous user rate 19,2 kbit/s | |
| **Comments:** | P104 TC11326 | |

| 110230 | Ref. to ETS 300 102-1 [1], subclauses 4.5.5, 4.5.18 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Successful/UDI/TC110230 | |
| **Selection criteria:** | V.110/X.30 rate adaption | |
| **Test purpose:** | Support of terminal adaptors V.110/X.30: Ensure that BC and LLC information is transported transparently through the network and correctly delivered to the called user. | |
| **Parameter values:** | BC=LLC=V110/X30 asynchronous user rate 2,4 kbit/s | |
| **Comments:** | | |

| 110231 | Ref. to ETS 300 102-1 [1], subclauses 4.5.5, 4.5.18 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Successful/UDI/TC110231 | |
| **Selection criteria:** | V.110/X.30 rate adaption | |
| **Test purpose:** | Support of terminal adaptors V.110/X.30: Ensure that BC and LLC information is transported transparently through the network and correctly delivered to the called user. | |
| **Parameter values:** | BC=LLC=V110/X30 asynchronous user rate 9,6 kbit/s | |
| **Comments:** | | |

| 110232 | Ref. to ETS 300 102-1 [1], subclauses 4.5.5, 4.5.18 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Successful/UDI/TC110232 | |
| **Selection criteria:** | V.110/X.30 rate adaption | |
| **Test purpose:** | Support of terminal adaptors V.110/X.30: Ensure that BC and LLC information is transported transparently through the network and correctly delivered to the called user. | |
| **Parameter values:** | BC=LLC=V110/X30 asynchronous user rate 19,2 kbit/s | |
| **Comments:** | | |

| 110233 | Ref. to ETS 300 102-1 [1], subclauses 4.5.5, 4.5.18 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Successful/UDI/TC110232 | |
| **Selection criteria:** | V.110/X.30 rate adaption | |
| **Test purpose:** | Support of terminal adaptors V.110/X.30: Ensure that the BC information is transported transparently through the network and correctly delivered to the called user. | |
| **Parameter values:** | BC=V110/X30 synchronous user rate 56 kbit/s, no LLC | |
| **Comments:** | | |

| 110234 | Ref. to ETS 300 102-1 [1], subclauses 4.5.5, 4.5.18 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Successful/UDI/TC110234 | |
| **Selection criteria:** | V.110/X.30 rate adaption | |
| **Test purpose:** | Support of terminal adaptors V.110/X.30: Ensure that the LLC information is transported transparently through the network and correctly delivered to the called user. | |
| **Parameter values:** | BC=UDI, LLC=V110/X30 synchronous user rate 56 kbit/s | |
| **Comments:** | | |

| 110235 | Ref. to ETS 300 102-1 [1], subclauses 4.5.5, 4.5.18 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Successful/UDI/TC110235 | |
| **Selection criteria:** | V.110/X.30 rate adaption | |
| **Test purpose:** | Support of terminal adaptors V.110/X.30: Ensure that BC and LLC information is transported transparently through the network and correctly delivered to the called user. | |
| **Parameter values:** | BC=LLC=V110/X30 synchronous user rate 56 kbit/s | |
| **Comments:** | | |

| 110240 | Ref. to ETS 300 102-1 [1], subclause 4.5.16 | Other relevant ref.: ETR 018 [28], subclause 7.1.3 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Successful/UDI/TC110240 | |
| **Selection criteria:** | Syntax-based videotex teleservice | |
| **Test purpose:** | Support of syntax-based videotex teleservice using end-to-end circuit connection: Ensure that the LLC and HLC information is transported transparently through the network and correctly delivered to the called user | |
| **Parameter values:** | BC=UDI, HLC=syntax-based videotex, LLC=telematic_term | |
| **Comments:** | P104 TC11309 | |

| 110250 | Ref. to ETS 300 102-1 [1], subclause 4.5.16 | Other relevant ref.: ETR 018 [28], subclause 6.3.7 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Successful/UDI/TC110250 | |
| **Selection criteria:** | FTAM teleservice | |
| **Test purpose:** | Support of file transfer & access management (FTAM) teleservice: Ensure that the LLC and HLC information is transported transparently through the network and correctly delivered to the called user | |
| **Parameter values:** | BC=UDI, HLC=FTAM, LLC=telematic_term | |
| **Comments:** | P104 TC11309 | |

| 110251 | Ref. to ETS 300 102-1 [1], subclause 4.5.16 | Other relevant ref.: ETR 018 [28], subclause 6.3.8 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Successful/UDI/TC110251 | |
| **Selection criteria:** | Eurofile transfer teleservice | |
| **Test purpose:** | Support of Eurofile transfer teleservice: Ensure that the LLC and HLC information is transported transparently through the network and correctly delivered to the called user | |
| **Parameter values:** | BC=UDI, HLC=Eurofile, LLC=telematic_term | |
| **Comments:** | P104 TC11309 | |

| 110301 | Ref. to ETS 300 102-1 [1], subclause 5.1.5.1 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Successful/Audio/TC110301 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that call establishment using en-bloc sending is performed correctly | |
| **Parameter values:** | BC=3,1 kHz audio, no HLC | |
| **Comments:** | P104 TC11201 | |

| 110302 | Ref. to ETS 300 102-1 [1], subclause 5.1.5.2 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Successful/Audio/TC110302 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that call establishment using overlap sending is performed correctly | |
| **Parameter values:** | BC=3,1 kHz audio, no HLC | |
| **Comments:** | P104 TC11202 | |

| 110303 | Ref. to ETS 300 102-1 [1], subclause 5.3.3 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Successful/Audio/TC110303 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that the call clearing procedure is performed correctly when the calling user clears after answer | |
| **Parameter values:** | BC=3,1 kHz audio, no HLC | |
| **Comments:** | P104 TC11203 | |

| 110304 | Ref. to ETS 300 102-1 [1], subclause 5.3.3 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Successful/Audio/TC110304 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that the call clearing procedure is performed correctly when the called user clears after answer | |
| **Parameter values:** | BC=3,1 kHz audio, no HLC | |
| **Comments:** | P104 TC11204 | |

| 110305 | Ref. to ETS 300 102-1 [1], subclause 4.5.16 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Successful/Audio/TC110305 | |
| **Selection criteria:** | Telefax G2/G3 terminals | |
| **Test purpose:** | Support of Telefax G2/G3: Ensure that the HLC information is transported transparently through the network and correctly delivered to the called user. | |
| **Parameter values:** | BC=3,1 kHz audio, HLC=facsimile group 2/3 | |
| **Comments:** | P104 TC11205 | |

| 110306 | Ref. to ETS 300 102-1 [1], subclause 4.5.18 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Successful/Audio/TC110306 | |
| **Selection criteria:** | Voice band data via modem | |
| **Test purpose:** | Support of voice band data via modem: Ensure that the LLC information is transported transparently through the network and correctly delivered to the called user. | |
| **Parameter values:** | BC=3,1 kHz audio, LLC = voice band data via modem | |
| **Comments:** | P104 TC11206 | |

| 110401 | Ref. to ETS 300 102-1 [1], subclause 5.1.5.1 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Successful/UDI-TA/TC110401 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that call establishment using en-bloc sending is performed correctly | |
| **Parameter values:** | BC=UDI/TA, no HLC | |
| **Comments:** | P104 TC11401 | |

| 110402 | Ref. to ETS 300 102-1 [1], subclause 5.1.5.2 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Successful/UDI-TA/TC110402 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that call establishment using overlap sending is performed correctly | |
| **Parameter values:** | BC=UDI/TA, no HLC | |
| **Comments:** | P104 TC11402 | |

| 110403 | Ref. to ETS 300 102-1 [1], subclause 5.3.3 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Successful/UDI-TA/TC110403 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that the call clearing procedure is performed correctly when the calling user clears after answer | |
| **Parameter values:** | BC=UDI/TA, no HLC | |
| **Comments:** | P104 TC11403 | |

| 110404 | Ref. to ETS 300 102-1 [1], subclause 5.3.3 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Successful/UDI-TA/TC110404 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that the call clearing procedure is performed correctly when the called user clears after answer | |
| **Parameter values:** | BC=UDI/TA, no HLC | |
| **Comments:** | P104 TC11404 | |

| 110405 | Ref. to ETS 300 267-1 [4], subclause 6.5.1 | Other relevant ref.: ETR 018 [28], subclause 6.3.4 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Successful/UDI-TA/TC110405 | |
| **Selection criteria:** | Telephony UDI-TA teleservice | |
| **Test purpose:** | Support of telephony UDI-TA teleservice: Ensure that the HLC information is transported transparently through the network and correctly delivered to the called user. | |
| **Parameter values:** | BC=UDI/TA, HLC=telephony,<br>telephony 7 kHz fallback not allowed SETUP message | |
| **Comments:** | P104 TC11405 | |

| 110406 | Ref. to ETS 300 267-1 [4], subclause 7.5.1 | Other relevant ref.: ETR 018 [28], subclause 6.3.5 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Successful/UDI-TA/TC110406 | |
| **Selection criteria:** | Videotelephony teleservice | |
| **Test purpose:** | Support of videotelephony teleservice: Ensure that the HLC information is transported transparently through the network and correctly delivered to the called user. | |
| **Parameter values:** | BC=UDI/TA, HLC = videotelephony_ic<br>videotelephony fallback not allowed SETUP message | |
| **Comments:** | | |

| 120101 | Ref. to ETS 300 102-1 [1], subclauses 5.1.4, G.1.1 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Unsuccessful/Speech/TC120101 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that, when calling to unallocated number, the network initiate call clearing to the calling user with cause value #1 "unassigned number" | |
| **Parameter values:** | BC=speech | |
| **Comments:** | P104 TC12001 | |

| 120102 | Ref. to ETS 300 102-1 [1], subclauses 5.2.5.1, G.1.7 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Unsuccessful/Speech/TC120102 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that, when the called user is busy and responds with a RELEASE COMPLETE message indicating cause value #17 "user busy", the network transport the cause value to the calling user | |
| **Parameter values:** | BC=speech | |
| **Comments:** | P104 TC12002 | |

| 120103 | Ref. to ETS 300 102-1 [1], subclauses 5.2.5.4, G.1.8 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Unsuccessful/Speech/TC120103 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that when the called user is not responding, the network initiate call clearing to the calling user with cause value #18 "no user responding" | |
| **Parameter values:** | BC=speech | |
| **Comments:** | P104 TC12003 | |

| 120104 | Ref. to ETS 300 102-1 [1], subclauses 5.2.5.4, G.1.9 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Unsuccessful/Speech/TC120104 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that when no answer from the called user (but user alerted), the network initiate call clearing to the calling user and called user with cause value #19 "no user responding (user alerted)" | |
| **Parameter values:** | BC=speech | |
| **Comments:** | P104 TC12004 | |

| 120105 | Ref. to ETS 300 102-1 [1], subclauses 5.1.9, 5.3.2, G.1.10 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Unsuccessful/Speech/TC120105 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that when the called user rejects the call and responds with a RELEASE COMPLETE message indicating cause value #21 "call rejected", the network transport the cause value to the calling user | |
| **Parameter values:** | BC=speech | |
| **Comments:** | P104 TC12005 | |

| 120106 | Ref. to ETS 300 102-1 [1], subclause G.1.13 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Unsuccessful/Speech/TC120106 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that when the called user terminal is not connected, the network initiate call clearing to the calling user with cause value #27 "destination out of order" | |
| **Parameter values:** | BC=speech | |
| **Comments:** | P104 TC12006 | |

| 120107 | Ref. to ETS 300 102-1 [1], subclauses 5.2.2, G.5.7 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Unsuccessful/Speech/TC120107 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that when the called user is not compatible and responds with a RELEASE COMPLETE message indicating cause value #88 "called user not compatible", the network transport the cause value to the calling user | |
| **Parameter values:** | BC=speech | |
| **Comments:** | P104 TC12009 | |

| 120108 | Ref. to ETS 300 102-1 [1], subclause G.1.6 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Unsuccessful/Speech/TC120108 | |
| **Selection criteria:** | Multipoint configuration for the called side | |
| **Test purpose:** | Ensure that when the calling user clears with cause value #16 "normal call clearing" before answer from called user, the network transport the cause value to the called user | |
| **Parameter values:** | BC=speech | |
| **Comments:** | | |

| 120109 | Ref. to ETS 300 102-1 [1], subclause G.1.6 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Unsuccessful/Speech/TC120109 | |
| **Selection criteria:** | Point-to-point configuration for the called side | |
| **Test purpose:** | Ensure that when the calling user clears with cause value #16 "normal call clearing" before answer from called user, the network transport the cause value to the called user | |
| **Parameter values:** | BC=speech | |
| **Comments:** | P104 TC12010 | |

| 120201 | Ref. to ETS 300 102-1 [1], subclauses 5.1.4, G.1.1 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Unsuccessful/UDI/TC120201 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that, when calling to unallocated number, the network initiate call clearing to the calling user with cause value #1 "unassigned number" | |
| **Parameter values:** | BC=UDI | |
| **Comments:** | | |

| 120202 | Ref. to ETS 300 102-1 [1], subclauses 5.2.5.1, G.1.7 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Unsuccessful/UDI/TC120202 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that, when the called user is busy and responds with a RELEASE COMPLETE message indicating cause value #17 "user busy", the network transport the cause value to the calling user | |
| **Parameter values:** | BC=UDI | |
| **Comments:** | | |

| 120203 | Ref. to ETS 300 102-1 [1], subclauses 5.2.5.4, G.1.8 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Unsuccessful/UDI/TC120203 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that, when the called user is not responding, the network initiate call clearing to the calling user with cause value #18 "no user responding" | |
| **Parameter values:** | BC=UDI | |
| **Comments:** | | |

| 120204 | Ref. to ETS 300 102-1 [1], subclauses 5.2.5.4, G.1.9 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Unsuccessful/UDI/TC120204 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that when no answer from the called user (but user alerted), the network initiate call clearing to the calling user and called user with cause value #19 "no user responding (user alerted)" | |
| **Parameter values:** | BC=UDI | |
| **Comments:** | | |

| 120205 | Ref. to ETS 300 102-1 [1], subclauses 5.1.9, 5.3.2, G.1.10 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Unsuccessful/UDI/TC120205 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that when the called user rejects the call and responds with a RELEASE COMPLETE message indicating cause value #21 "call rejected", the network transport the cause value to the calling user | |
| **Parameter values:** | BC=UDI | |
| **Comments:** | | |

| 120206 | Ref. to ETS 300 102-1 [1], subclause G.1.13 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Unsuccessful/UDI/TC120206 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that when the called user terminal is not connected, the network initiate call clearing to the calling user with cause value #27 "destination out of order" | |
| **Parameter values:** | BC=UDI | |
| **Comments:** | | |

| 120207 | Ref. to ETS 300 102-1 [1], subclauses 5.2.2, G.5.7 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Unsuccessful/UDI/TC120207 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that when the called user is not compatible and responds with a RELEASE COMPLETE message indicating cause value #88 "called user not compatible", the network transport the cause value to the calling user | |
| **Parameter values:** | BC=UDI | |
| **Comments:** | | |

| 120208 | Ref. to ETS 300 102-1 [1], subclause G.1.6 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Unsuccessful/UDI/TC120208 | |
| **Selection criteria:** | Multipoint configuration for the called side | |
| **Test purpose:** | Ensure that when the calling user clears with cause value #16 "normal call clearing" before answer from called user, the network transport the cause value to the called user | |
| **Parameter values:** | BC=UDI | |
| **Comments:** | | |

| 120209 | Ref. to ETS 300 102-1 [1], subclause G.1.6 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Unsuccessful/UDI/TC120209 | |
| **Selection criteria:** | Point-to-point configuration for the called side | |
| **Test purpose:** | Ensure that when the calling user clears with cause value #16 "normal call clearing" before answer from called user, the network transport the cause value to the called user | |
| **Parameter values:** | BC=UDI | |
| **Comments:** | | |

| 120301 | Ref. to ETS 300 102-1 [1], subclauses 5.1.4, G.1.1 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Unsuccessful/Audio/TC120301 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that, when calling to unallocated number, the network initiate call clearing to the calling user with cause value #1 "unassigned number" | |
| **Parameter values:** | BC=3,1 kHz audio | |
| **Comments:** | | |

| 120302 | Ref. to ETS 300 102-1 [1], subclauses 5.2.5.1, G.1.7 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Unsuccessful/Audio/TC120302 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that, when the called user is busy and responds with a RELEASE COMPLETE message indicating cause value #17 "user busy", the network transport the cause value to the calling user | |
| **Parameter values:** | BC=3,1 kHz audio | |
| **Comments:** | | |

| 120303 | Ref. to ETS 300 102-1 [1], subclauses 5.2.5.4, G.1.8 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Unsuccessful/Audio/TC120303 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that, when the called user is not responding, the network initiate call clearing to the calling user with cause value #18 "no user responding" | |
| **Parameter values:** | BC=3,1 kHz audio | |
| **Comments:** | | |

| 120304 | Ref. to ETS 300 102-1 [1], subclauses 5.2.5.4, G.1.9 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Unsuccessful/Audio/TC120304 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that when no answer from the called user (but user alerted), the network initiate call clearing to the calling user and called user with cause value #19 "no user responding (user alerted)" | |
| **Parameter values:** | BC=3,1 kHz audio | |
| **Comments:** | | |

| 120305 | Ref. to ETS 300 102-1 [1], subclauses 5.1.9, 5.3.2, G.1.10 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Unsuccessful/Audio/TC120305 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that when the called user rejects the call and responds with a RELEASE COMPLETE message indicating cause value #21 "call rejected", the network transport the cause value to the calling user | |
| **Parameter values:** | BC=3,1 kHz audio | |
| **Comments:** | | |

| 120306 | Ref. to ETS 300 102-1 [1], subclause G.1.13 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Unsuccessful/Audio/TC120306 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that when the called user terminal is not connected, the network initiate call clearing to the calling user with cause value #27 "destination out of order" | |
| **Parameter values:** | BC=3,1 kHz audio | |
| **Comments:** | | |

| 120307 | Ref. to ETS 300 102-1 [1], subclauses 5.2.2, G.5.7 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Unsuccessful/Audio/TC120307 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that when the called user is not compatible and responds with a RELEASE COMPLETE message indicating cause value #88 "called user not compatible", the network transport the cause value to the calling user | |
| **Parameter values:** | BC=3,1 kHz audio | |
| **Comments:** | | |

| 120308 | Ref. to ETS 300 102-1 [1], subclause G.1.6 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Unsuccessful/Audio/TC120308 | |
| **Selection criteria:** | Multipoint configuration for the called side | |
| **Test purpose:** | Ensure that when the calling user clears with cause value #16 "normal call clearing" before answer from called user, the network transport the cause value to the called user | |
| **Parameter values:** | BC=3,1 kHz audio | |
| **Comments:** | | |

| 120309 | Ref. to ETS 300 102-1 [1], subclause G.1.6 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Unsuccessful/Audio/TC120309 | |
| **Selection criteria:** | Point-to-point configuration for the called side | |
| **Test purpose:** | Ensure that when the calling user clears with cause value #16 "normal call clearing" before answer from called user, the network transport the cause value to the called user | |
| **Parameter values:** | BC=3,1 kHz audio | |
| **Comments:** | | |

| 120401 | Ref. to ETS 300 102-1 [1], subclauses 5.1.4, G.1.1 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Unsuccessful/UDI-TA/TC120401 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that, when calling to unallocated number, the network initiate call clearing to the calling user with cause value #1 "unassigned number" | |
| **Parameter values:** | BC=UDI/TA | |
| **Comments:** | | |

| 120402 | Ref. to ETS 300 102-1 [1], subclauses 5.2.5.1, G.1.7 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Unsuccessful/UDI-TA/TC120402 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that, when the called user is busy and responds with a RELEASE COMPLETE message indicating cause value #17 "user busy", the network transport the cause value to the calling user | |
| **Parameter values:** | BC=UDI/TA | |
| **Comments:** | | |

| 120403 | Ref. to ETS 300 102-1 [1], subclauses 5.2.5.4, G.1.8 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Unsuccessful/UDI-TA/TC120403 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that, when the called user is not responding, the network initiate call clearing to the calling user with cause value #18 "no user responding" | |
| **Parameter values:** | BC=UDI/TA | |
| **Comments:** | | |

| 120404 | Ref. to ETS 300 102-1 [1], subclauses 5.2.5.4, G.1.9 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Unsuccessful/UDI-TA/TC120404 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that when no answer from the called user (but user alerted), the network initiate call clearing to the calling user and called user with cause value #19 "no user responding (user alerted)" | |
| **Parameter values:** | BC=UDI/TA | |
| **Comments:** | | |

| 120405 | Ref. to ETS 300 102-1 [1], subclauses 5.1.9, 5.3.2, G.1.10 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Unsuccessful/UDI-TA/TC120405 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that when the called user rejects the call and responds with a RELEASE COMPLETE message indicating cause value #21 "call rejected", the network transport the cause value to the calling user | |
| **Parameter values:** | BC=UDI/TA | |
| **Comments:** | | |

| 120406 | Ref. to ETS 300 102-1 [1], subclause G.1.13 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Unsuccessful/UDI-TA/TC120406 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that when the called user terminal is not connected, the network initiate call clearing to the calling user with cause value #27 "destination out of order" | |
| **Parameter values:** | BC=UDI/TA | |
| **Comments:** | | |

| 120407 | Ref. to ETS 300 102-1 [1], subclauses 5.2.2, G.5.7 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Unsuccessful/UDI-TA/TC120407 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that when the called user is not compatible and responds with a RELEASE COMPLETE message indicating cause value #88 "called user not compatible", the network transport the cause value to the calling user | |
| **Parameter values:** | BC=UDI/TA | |
| **Comments:** | | |

| 120408 | Ref. to ETS 300 102-1 [1], subclause G.1.6 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Unsuccessful/UDI-TA/TC120408 | |
| **Selection criteria:** | Multipoint configuration for the called side | |
| **Test purpose:** | Ensure that when the calling user clears with cause value #16 "normal call clearing" before answer from called user, the network transport the cause value to the called user | |
| **Parameter values:** | BC=UDI/TA | |
| **Comments:** | | |

| 120409 | Ref. to ETS 300 102-1 [1], subclause G.1.6 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Basic_call/Unsuccessful/UDI-TA/TC120409 | |
| **Selection criteria:** | Point-to-point configuration for the called side | |
| **Test purpose:** | Ensure that when the calling user clears with cause value #16 "normal call clearing" before answer from called user, the network transport the cause value to the called user | |
| **Parameter values:** | BC=UDI/TA | |
| **Comments:** | | |

#### 6.2.1.2 Test purposes for ISDN-ISDN, supplementary services

##### 6.2.1.2.1 Bearer capability "speech"

| 210101 | Ref. to ETS 300 092-1 [7], subclause 9.3 | Other relevant ref.: ETS 300 102-1 [1], subclauses 4.5.10, 4.5.11 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/CLIP/TC210101 | |
| **Selection criteria:** | The called user is provided with CLIP | |
| **Test purpose:** | Ensure that when Calling party number is provided by the calling user, Type of number "subscriber number", with Calling party subaddress, the Calling party number and Calling party subaddress information elements are correctly delivered to the called (served) user. | |
| **Parameter values:** | BC=speech, SI=UPVP, TON=international | |
| **Comments:** | | |

| 210102 | Ref. to ETS 300 092-1 [7], subclause 9.3 | Other relevant ref.: ETS 300 102-1 [1], subclauses 4.5.10, 4.5.11 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/CLIP/TC210102 | |
| **Selection criteria:** | The called user is provided with CLIP | |
| **Test purpose:** | Ensure that when Calling party number is provided by the calling user, Type of number "national number", with Calling party subaddress, the Calling party number and Calling party subaddress information elements are correctly delivered to the called (served) user. | |
| **Parameter values:** | BC=speech, SI=UPVP, TON=international | |
| **Comments:** | | |

| 210103 | Ref. to ETS 300 092-1 [7], subclause 9.3 | Other relevant ref.: ETS 300 102-1 [1], subclauses 4.5.10, 4.5.11 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/CLIP/TC210103 | |
| **Selection criteria:** | The called user is provided with CLIP | |
| **Test purpose:** | Ensure that when Calling party number is provided by the calling user, Type of number "international number", with Calling party subaddress, the Calling party number and Calling party subaddress information elements are correctly delivered to the called (served) user. | |
| **Parameter values:** | BC=speech, SI=UPVP, TON=international | |
| **Comments:** | | |

| 210104 | Ref. to ETS 300 092-1 [7], subclause 9.3 | Other relevant ref.: ETS 300 102-1 [1], subclauses 4.5.10, 4.5.11 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/CLIP/TC210104 | |
| **Selection criteria:** | The called user is provided with CLIP | |
| **Test purpose:** | Ensure that when Calling party number is provided by the calling user, Type of number "unknown", with Calling party subaddress, the Calling party number and Calling party subaddress information elements are correctly delivered to the called (served) user. | |
| **Parameter values:** | BC=speech, SI=UPVP, TON=international | |
| **Comments:** | P104 TC20101* | |

| 210105 | Ref. to ETS 300 092-1 [7], subclause 9.3 | Other relevant ref.: ETS 300 102-1 [1], subclause 4.5.10 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/CLIP/TC210105 | |
| **Selection criteria:** | The called user is provided with CLIP | |
| **Test purpose:** | Ensure that when no Calling party number information element is provided by the calling user, (and no Calling party subaddress), the Calling party number information element is network provided and correctly delivered to the called (served) user. | |
| **Parameter values:** | BC=speech, SI=NP, TON=international | |
| **Comments:** | P104 TC20102* | |

| 210201 | Ref. to ETS 300 093-1 [8], subclause 9.4.1 | Other relevant ref.: ETS 300 092-1 [7], subclause A.2, figure 2 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/CLIR/TC210201 | |
| **Selection criteria:** | The calling user is provided with CLIR permanent mode subscription, the called user with CLIP | |
| **Test purpose:** | Ensure that when the Calling party number is provided by the calling user, with Calling party subaddress, the Calling party number information element is delivered to the called user without any digit information. The Calling party subaddress shall not be present | |
| **Parameter values:** | BC=speech, PI=PR, SI=NP, TON=unknown, NPI=unknown | |
| **Comments:** | P104 TC20201* | |

| 210202 | Ref. to ETS 300 093-1 [8], subclause 9.4.1 | Other relevant ref.: ETS 300 092-1 [7], subclause A.2, figure 2 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/CLIR/TC210202 | |
| **Selection criteria:** | The calling user is provided with CLIR permanent mode subscription, the called user with CLIP | |
| **Test purpose:** | Ensure that when no Calling party number is provided by the calling user (and no Calling party subaddress), the Calling party number information element is network provided and delivered to the called user without any digit information. | |
| **Parameter values:** | BC=speech, PI=PR, SI=NP, TON=unknown, NPI=unknown | |
| **Comments:** | | |

| 210301 | Ref. to ETS 300 097-1 [9], subclause 9.5.1 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/COLP/TC210301 | |
| **Selection criteria:** | The calling user is provided with COLP | |
| **Test purpose:** | Ensure that when the Connected number is provided by the called user, Type of number "subscriber number", with Connected subaddress, the Connected number and Connected subaddress information elements are correctly delivered to the calling (served) user. | |
| **Parameter values:** | BC=speech, SI=UPVP, TON=international | |
| **Comments:** | | |

| 210302 | Ref. to ETS 300 097-1 [9], subclause 9.5.1 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/COLP/TC210302 | |
| **Selection criteria:** | The calling user is provided with COLP | |
| **Test purpose:** | Ensure that when the Connected number is provided by the called user, Type of number "national number", with Connected subaddress, the Connected number and Connected subaddress information elements are correctly delivered to the calling (served) user. | |
| **Parameter values:** | BC=speech, SI=UPVP, TON=international | |
| **Comments:** | | |

| 210303 | Ref. to ETS 300 097-1 [9], subclause 9.5.1 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/COLP/TC210303 | |
| **Selection criteria:** | The calling user is provided with COLP | |
| **Test purpose:** | Ensure that when the Connected number is provided by the called user, Type of number "international number", with Connected subaddress, the Connected number and Connected subaddress information elements are correctly delivered to the calling (served) user. | |
| **Parameter values:** | BC=speech, SI=UPVP, TON=international | |
| **Comments:** | | |

| 210304 | Ref. to ETS 300 097-1 [9], subclause 9.5.1 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/COLP/TC210304 | |
| **Selection criteria:** | The calling user is provided with COLP | |
| **Test purpose:** | Ensure that when the Connected number is provided by the called user, Type of number "unknown", with Connected subaddress, the Connected number and Connected subaddress information elements are correctly delivered to the calling (served) user. | |
| **Parameter values:** | BC=speech, SI=UPVP, TON=international | |
| **Comments:** | P104 TC20301* | |

| 210305 | Ref. to ETS 300 097-1 [9], subclause 9.5.1 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/COLP/TC210305 | |
| **Selection criteria:** | Calling user is provided with COLP | |
| **Test purpose:** | Ensure that when no Connected number is provided by the called user (and no Connected subaddress), the Connected number information element is network provided and correctly delivered to the calling (served) user. | |
| **Parameter values:** | BC=speech, SI=NP, TON=international | |
| **Comments:** | P104 TC20302* | |

| 210401 | Ref. to ETS 300 098-1 [10], subclauses 9.3.1, 9.4.1 | Other relevant ref.: ETS 300 097-1 [9], subclause A.1, figure 4 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/COLR/TC210401 | |
| **Selection criteria:** | The called (served) user is provided with COLR permanent mode subscription, the calling user with COLP | |
| **Test purpose:** | Ensure that when the Connected number is provided by the called user, with Connected subaddress, the Connected number information element is delivered to the calling user without any digit information. The Connected subaddress shall not be present | |
| **Parameter values:** | BC=speech, PI=PR, SI=NP, TON=unknown, NPI=unknown | |
| **Comments:** | P104 TC20401* | |

| 210402 | Ref. to ETS 300 098-1 [10], subclauses 9.3.1, 9.4.1 | Other relevant ref.: ETS 300 097-1 [9], subclause A.2, figure 4 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/COLR/TC210402 | |
| **Selection criteria:** | The called (served) user is provided with COLR permanent mode subscription, the calling user with COLP | |
| **Test purpose:** | Ensure that when no Connected number is provided by the called user (and no Connected subaddress), the Connected number information element is network provided and delivered to the calling user without any digit information. | |
| **Parameter values:** | BC=speech, PI=PR, SI=NP, TON=unknown, NPI=unknown | |
| **Comments:** | | |

| 210501 | Ref. to ETS 300 138-1 [11], subclauses 9.2.2, 9.2.4 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/CUG/TC210501 | |
| **Selection criteria:** | Calling user and called user belong to the same (international) CUG | |
| **Test purpose:** | Ensure that when the calling user and called user belongs to the same (international) CUG, call establishment is possible | |
| **Parameter values:** | BC=speech, CUG default request | |
| **Comments:** | P104 TC20501 | |

| 210502 | Ref. to ETS 300 138-1 [11], subclauses 9.2.2, 9.2.4 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/CUG/TC210502 | |
| **Selection criteria:** | Calling user belongs to a CUG with outgoing access allowed and called user belongs to another CUG with incoming access allowed | |
| **Test purpose:** | Ensure that when the calling user belongs to a CUG with outgoing access allowed and the called user belongs to another CUG with incoming access allowed, the call establishment is possible | |
| **Parameter values:** | BC=speech, CUG default request | |
| **Comments:** | P104 TC20502 | |

| 210503 | Ref. to ETS 300 138-1 [11], subclause 9.2.2 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/CUG/TC210503 | |
| **Selection criteria:** | Calling user belongs to a CUG with outgoing access "allowed" and the called user is not a CUG subscriber | |
| **Test purpose:** | Ensure that when the calling user belongs to a CUG with outgoing access allowed and the called user is not a CUG subscriber, the call establishment is possible | |
| **Parameter values:** | BC=speech, CUG default request | |
| **Comments:** | P104 TC20503 | |

| 210504 | Ref. to ETS 300 138-1 [11], subclause 9.2.3 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/CUG/TC210504 | |
| **Selection criteria:** | Called user belong to a CUG with incoming access "not allowed" and calling user is not member of CUG | |
| **Test purpose:** | Ensure that when the called user belongs to a CUG with incoming access "not allowed" and the calling user is not member of CUG, call establishment is not possible and the network initiate call clearing to the calling user with cause value #87 "user not a member of CUG". | |
| **Parameter values:** | BC=speech | |
| **Comments:** | P104 TC12007 | |

| 210505 | Ref. to ETS 300 138-1 [11], subclause 9.2.2 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/CUG/TC210505 | |
| **Selection criteria:** | Calling user belong to a CUG with outgoing access "not allowed" and called user is not member of CUG | |
| **Test purpose:** | Ensure that when the calling user belong to CUG with outgoing access "not allowed" and the called user is not member of CUG, call establishment is not possible and the network initiate call clearing to the calling user with cause value #87 "user not a member of CUG". | |
| **Parameter values:** | BC=speech, CUG default request | |
| **Comments:** | P104 TC12008 | |

| 210601 | Ref. to ETS 300 061-1 [12], subclause 9.2 | Other relevant ref.: ETS 300 102-1 [1], subclause 4.5.9 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/SUB/TC220601 | |
| **Selection criteria:** | The called (served) user is provided with SUB | |
| **Test purpose:** | Ensure that when the Called party subaddress is provided by the calling user, the Called party subaddress is correctly delivered to the called (served) user | |
| **Parameter values:** | BC=speech | |
| **Comments:** | P104 TC20601 | |

| 210602 | Ref. to ETS 300 061-1 [12], subclause 9.2 | Other relevant ref.: ETS 300 102-1 [1], subclause 4.5.9 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/SUB/TC210602 | |
| **Selection criteria:** | The called (served) user is provided with SUB | |
| **Test purpose:** | Ensure that when the Called party subaddress is provided by the calling user with length=minimum, the Called party subaddress is correctly delivered to the called (served) user without any digit information | |
| **Parameter values:** | BC=speech | |
| **Comments:** | | |

| 210701 | Ref. to ETS 300 055-1 [13], subclause 9.2.1 | Other relevant ref.: ETS 300 102-1 [1], subclause 5.6 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/TP/TC210701 | |
| **Selection criteria:** | The calling user shall be a basic access | |
| **Test purpose:** | Ensure that the called user is notified of the call suspension and resumption by the calling user (no call identity is used) | |
| **Parameter values:** | BC=speech | |
| **Comments:** | P104 TC20701 | |

| 210702 | Ref. to ETS 300 055-1 [13], subclause 9.2.1 | Other relevant ref.: ETS 300 102-1 [1], subclause 5.6 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/TP/TC210702 | |
| **Selection criteria:** | The called user shall be a basic access | |
| **Test purpose:** | Ensure that the calling user is notified of the call suspension and resumption by the called user (no call identity is used) | |
| **Parameter values:** | BC=speech | |
| **Comments:** | | |

| 210703 | Ref. to ETS 300 055-1 [13], subclause 9.2.2 | Other relevant ref.: ETS 300 102-1 [1], subclause 5.6.5 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/TP/TC210703 | |
| **Selection criteria:** | The calling user shall be a basic access | |
| **Test purpose:** | Ensure that when the call is suspended, with the expiry of timer T307 before the call re-establishment, the network starts call clearing to the (still) active side with cause value #102 "recovery on timer expiry". | |
| **Parameter values:** | BC=speech | |
| **Comments:** | | |

| 210801 | Ref. to ETS 300 286-1 [14], subclauses 9.1.1.1, 9.1.2.1 | Other relevant ref.: ETS 300 102-1 [1], subclause 4.5.29 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/UUS1/TC210801 | |
| **Selection criteria:** | The calling (served) user is provided with UUS1 implicit request | |
| **Test purpose:** | Ensure that the network can transport a User-user information element included in the SETUP message sent from the calling user and delivered in the SETUP message sent by the network to the called user | |
| **Parameter values:** | BC=speech, UI length=32 | |
| **Comments:** | P104 TC20801 | |

| 210802 | Ref. to ETS 300 286-1 [14], subclause 9.1.2.1 | Other relevant ref.: ETS 300 102-1 [1], subclause 4.5.29 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/UUS1/TC210802 | |
| **Selection criteria:** | The calling (served) user is provided with UUS1 implicit request | |
| **Test purpose:** | Ensure that after implicit activation of UUS1, the network can transport a User-user information element included in the ALERTING message sent from the called user to the calling user. | |
| **Parameter values:** | BC=speech, UI length=32 | |
| **Comments:** | P104 TC20802 | |

| 210803 | Ref. to ETS 300 286-1 [14], subclause 9.1.2.1 | Other relevant ref.: ETS 300 102-1 [1], subclause 4.5.29 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/UUS1/TC210803 | |
| **Selection criteria:** | The calling (served) user is provided with UUS1 implicit request | |
| **Test purpose:** | Ensure that after implicit activation of UUS1, the network can transport a User-user information element included in the CONNECT message sent from the called user to the calling user | |
| **Parameter values:** | BC=speech, UI length=32 | |
| **Comments:** | P104 TC20803 | |

| 210804 | Ref. to ETS 300 286-1 [14], subclause 9.1.2.2.1a | Other relevant ref.: ETS 300 102-1 [1] |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/UUS1/TC210804 | |
| **Selection criteria:** | The calling (served) user is provided with UUS1 implicit request | |
| **Test purpose:** | Ensure that after implicit activation of UUS1 and with the call in the active state, the network can transport a User-user information element included in a call clearing DISCONNECT message sent from the calling user and delivered in the DISCONNECT message sent by the network to the called user. | |
| **Parameter values:** | BC=speech, UI length=32 | |
| **Comments:** | P104 TC20804 | |

| 210805 | Ref. to ETS 300 286-1 [14], subclause 9.1.2.2.1b | Other relevant ref.: ETS 300 102-1 [1] |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/UUS1/TC210805 | |
| **Selection criteria:** | The calling (served) user is provided with UUS1 implicit request | |
| **Test purpose:** | Ensure that after implicit activation of UUS1, the network can transport a User-user information element included in premature clearing RELEASE COMPLETE message sent from the called user and delivered in the DISCONNECT message sent by the network to the calling user | |
| **Parameter values:** | BC=speech, UI length=32 | |
| **Comments:** | | |

| 210806 | Ref. to ETS 300 286-1 [14], subclause 9.1.1.1.1 | Other relevant ref.: ETS 300 102-1 [1], subclause 7.1.3.2 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/UUS1/TC210806 | |
| **Selection criteria:** | The calling (served) user is provided with UUS1 implicit request | |
| **Test purpose:** | Ensure that implicit activation of UUS1 with a User-user information element with the minimum length of three octets (without any user information), included in the SETUP message sent from the calling user, is supported. | |
| **Parameter values:** | BC=speech | |
| **Comments:** | | |

| 210807 | Ref. to ETS 300 286-1 [14], subclause 9.1.2.2.1b | Other relevant ref.: ETS 300 102-1 [1] |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/UUS1/TC210807 | |
| **Selection criteria:** | The calling (served) user is provided with UUS1 implicit request | |
| **Test purpose:** | Ensure that after implicit activation of UUS1 and with the call in the active state, the network can transport a User-user information element included in a call clearing DISCONNECT message sent from the called user and delivered in the DISCONNECT message sent by the network to the calling user. | |
| **Parameter values:** | BC=speech, UI length=32 | |
| **Comments:** | | |

| 210808 | Ref. to ETS 300 286-1 [14], subclause 9.1.2.2.1b | Other relevant ref.: ETS 300 102-1 [1] |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/UUS1/TC210808 | |
| **Selection criteria:** | The calling (served) user is provided with UUS1 implicit request<br>Multipoint configuration for the called side. | |
| **Test purpose:** | Ensure that after implicit activation of UUS1, the network can transport a User-user information element associated with the highest priority cause included in premature clearing RELEASE COMPLETE messages sent from the called users and delivered in the DISCONNECT message sent by the network to the calling user | |
| **Parameter values:** | BC=speech, UI length=32 | |
| **Comments:** | | |

| 210809 | Ref. to ETS 300 286-1 [14], subclause 9.1.2.2.1b | Other relevant ref.: ETS 300 102-1 [1] |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/UUS1/TC210809 | |
| **Selection criteria:** | The calling (served) user is provided with UUS1 implicit request. | |
| **Test purpose:** | Ensure that after implicit activation of UUS1, the network can transport a User-user information element included in a premature clearing DISCONNECT message sent from the called user and delivered in the DISCONNECT message sent by the network to the calling user | |
| **Parameter values:** | BC=speech, UI length=32 | |
| **Comments:** | | |

| 210810 | Ref. to ETS 300 286-1 [14], subclause 9.1.2.1 | Other relevant ref.: ETS 300 102-1 [1], subclause 4.5.29 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/UUS1/TC210810 | |
| **Selection criteria:** | The calling (served) user is provided with UUS1 implicit request.<br>Multipoint configuration for the called side. | |
| **Test purpose:** | Ensure that after implicit activation of UUS1, the network can transport a User-user information element included in the first received ALERTING message sent from the called users to the calling user in a ALERTING message | |
| **Parameter values:** | BC=speech, UI length=32 | |
| **Comments:** | P104 TC20802 | |

| 210811 | Ref. to ETS 300 286-1 [14], subclause 9.1.2.2.1b | Other relevant ref.: ETS 300 102-1 [1] |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/UUS1/TC210811 | |
| **Selection criteria:** | The calling (served) user is provided with UUS1 implicit request.<br>Multipoint configuration for the called side. | |
| **Test purpose:** | Ensure that after implicit activation of UUS1, the network can transport a User-user information element associated with the highest priority cause included in premature clearing DISCONNECT messages sent from the called users and delivered in the DISCONNECT message sent by the network to the calling user | |
| **Parameter values:** | BC=speech, UI length=32 | |
| **Comments:** | | |

| 210831 | Ref. to ETS 300 286-1 [14], subclause 9.1.1.2.2 | Other relevant ref.: ETS 300 102-1 [1], subclause 7.1.3.6 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/UUS1/TC210831 | |
| **Selection criteria:** | The calling (served) user is provided with UUS1 explicit request | |
| **Test purpose:** | Ensure that after explicit request of UUS1 indicating "required", if the called user rejects the call with a RELEASE COMPLETE message indicating cause value #29 "facility rejected", the network transport the cause value to the calling user. A UUS1 rejection with Error value "rejectedByUser" shall be included in the message | |
| **Parameter values:** | BC=speech | |
| **Comments:** | P104 TC12011 | |

| 210832 | Ref. to ETS 300 286-1 [14], subclause 9.1.1.2.2 | Other relevant ref.: ETS 300 102-1 [1], subclause 7.1.3.6 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/UUS1/TC210832 | |
| **Selection criteria:** | The calling (served) user is provided with UUS1 explicit request | |
| **Test purpose:** | Ensure that after explicit request of UUS1 indicating "required", if the destination network rejects the UUS1 request, the network initiates call clearing to the calling user with cause value #69 "requested facility not implemented" or cause value #29 "facility rejected". A UUS1 rejection with the Error value "rejectedByUser" shall be included in the message. | |
| **Parameter values:** | BC=speech | |
| **Comments:** | P104 TC12012* | |

| 210833 | Ref. to ETS 300 286-1 [14], subclause 9.1.1.2.1 | Other relevant ref.: ETS 300 102-1 [1], subclause 7.1.3.2 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/UUS1/TC210833 | |
| **Selection criteria:** | The calling (served) user is provided with UUS1 explicit request | |
| **Test purpose:** | Ensure that with the explicit request of UUS1 indicating "preferred", the network can transport a User-user information element included in the SETUP message sent from the calling user and delivered in the SETUP message sent by the network to the called user, and return the explicit service acceptance from the called user | |
| **Parameter values:** | BC=speech, UI length=32 | |
| **Comments:** | P104 TC20805 | |

| 210851 | Ref. to ETS 300 286-1 [14], subclause 9.2.2.1 | Other relevant ref.: ETS 300 102-1 [1], subclause 7.1.4.4 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/UUS2/TC210851 | |
| **Selection criteria:** | The calling (served) user is provided with UUS2.<br>Point-to-point configuration for the called side | |
| **Test purpose:** | Ensure that after activation of UUS2 indicating "preferred", the network can transport two USER INFORMATION messages in each direction, between the ALERTING and the CONNECT messages. | |
| **Parameter values:** | BC=speech, UI length=32 | |
| **Comments:** | P104 TC20806 | |

| 210852 | Ref. to ETS 300 286-1 [14], subclause 9.2.1.2 | Other relevant ref.: ETS 300 102-1 [1], subclause 7.1.4.3 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/UUS2/TC210852 | |
| **Selection criteria:** | The calling (served) user is provided with UUS2.<br>Point-to-point configuration for the called side. | |
| **Test purpose:** | Ensure that after activation of UUS2 indicating "preferred", if the network does not receive an explicit acceptance or rejection in the ALERTING message from the called user, a UUS2 rejection with the Error value "rejected by the user, it is returned to the calling user in a ALERTING message sent from the network and the call can be established. | |
| **Parameter values:** | BC=speech | |
| **Comments:** | P104 TC20807 | |

| 210871 | Ref. to ETS 300 286-1 [14], subclause 9.3.1.1 | Other relevant ref.: ETS 300 102-1 [1], subclause 7.1.5.2 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/UUS3/TC210871 | |
| **Selection criteria:** | The calling (served) user is provided with UUS3 | |
| **Test purpose:** | Ensure that after activation of UUS3 during call establishment indicating "preferred", the network can transport USER INFORMATION messages in both directions during the Active state of the call | |
| **Parameter values:** | BC=speech, UI length=32 | |
| **Comments:** | P104 TC20808 | |

| 210872 | Ref. to ETS 300 286-1 [14], subclause 9.3.1.1.2 | Other relevant ref.: ETS 300 102-1 [1], subclause 7.1.5.3 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/UUS3/TC210872 | |
| **Selection criteria:** | The calling (served) user is provided with UUS3 | |
| **Test purpose:** | Ensure that after the calling user request UUS3 during call establishment indicating "preferred", if the network does not receive an explicit acceptance or rejection in the CONNECT message from the called user, a UUS3 rejection with the Error value "rejected by the user" is included in the CONNECT message sent to the calling user. | |
| **Parameter values:** | BC=speech | |
| **Comments:** | P104 TC20809 | |

| 210873 | Ref. to ETS 300 286-1 [14], subclause 9.3.1.1 | Other relevant ref.: ETS 300 102-1 [1], subclause 7.1.5.2 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/UUS3/TC210873 | |
| **Selection criteria:** | The calling (served) user is provided with UUS3 | |
| **Test purpose:** | Ensure that after activation of UUS3 during call establishment indicating "required", the network can transport USER INFORMATION messages in both directions during the Active state of the call | |
| **Parameter values:** | BC=speech, UI length=32 | |
| **Comments:** | P104 TC20810 | |

| 210874 | Ref. to ETS 300 286-1 [14], subclause 9.3.1.2.1 | Other relevant ref.: ETS 300 102-1 [1], subclause 7.1.5.4 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/UUS3/TC210874 | |
| **Selection criteria:** | The calling (served) user is provided with UUS3 | |
| **Test purpose:** | Ensure that after activation of UUS3 during the Active call state indicating "preferred", the network can transport USER INFORMATION messages in both directions during the Active state of the call | |
| **Parameter values:** | BC=speech, UI length=32 | |
| **Comments:** | P104 TC20811 | |

| 210875 | Ref. to ETS 300 286-1 [14], subclause 9.3.1.2.2 | Other relevant ref.: ETS 300 102-1 [1], subclause 7.1.5.5 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/UUS3/TC210875 | |
| **Selection criteria:** | The calling (served) user is provided with UUS3 | |
| **Test purpose:** | Ensure that after the calling user request UUS3 during the Active call state indicating "preferred", if the called user rejects the service 3 request, the network can transport the FACILITY message including a UUS3 rejection with the Error value "rejected by the user" from the called user to the calling user. | |
| **Parameter values:** | BC=speech | |
| **Comments:** | P104 TC20812 | |

| 210901 | Ref. to ETS 300 185-1 [15], subclause 9.2.2 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/CONF/TC210901 | |
| **Selection criteria:** | The user A is in network N1 and is provided with CONF. The user B and user C are in network N2. | |
| **Test purpose:** | Ensure that user A can establish a conference call with user B and user C, and the remote parties are notified of the conference call progress related to Establish, Isolate, Reattach, Drop and Disconnect actions. | |
| **Parameter values:** | BC=speech | |
| **Comments:** | | |

| 211101 | Ref. to ETS 300 207-1 [16], subclause 9.2.2, 9.2.5 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/CFU/TC211101 | |
| **Selection criteria:** | The user A and the user C are in network N1. The user B is in network N2 and is provided with CFU ("calling user is notified of call diversion"=Yes, with diverted-to number, "diverting number is released to the diverted-to user"=Yes). | |
| **Test purpose:** | Ensure that when user A calls user B, the call is forwarded to user C, user A is notified of call diversion and informed of the diverted-to number (user C has presentation allowed - no COLR) and user C is informed of the forwarding number (user B has presentation allowed). | |
| **Parameter values:** | BC=speech, CF active | |
| **Comments:** | | |

| 211102 | Ref. to ETS 300 207-1 [16], subclauses 9.2.2, 9.2.5 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/CFU/TC211102 | |
| **Selection criteria:** | The user A and the user C are in network N1 and user C is provided with COLR. The user B is in network N2 and is provided with CFU ("calling user is notified of call diversion"=Yes, with diverted-to number, "diverting number is released to the diverted-to user"=No). | |
| **Test purpose:** | Ensure that when user A calls user B, the call is forwarded to user C, user A is notified of call diversion and not informed of the diverted-to number (user C has presentation not allowed - COLR) and user C is not informed of the forwarding number (user B has presentation not allowed). | |
| **Parameter values:** | BC=speech, CF active | |
| **Comments:** | | |

| 211201 | Ref. to ETS 300 207-1 [16], subclauses 9.2.2, 9.2.4.3, 9.2.5 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/CFB/TC211201 | |
| **Selection criteria:** | The user A and the user C are in network N1. The user B is in network N2 and is provided with CFB ("calling user is notified of call diversion"=Yes, with diverted-to number, "diverting number is released to the diverted-to user"=Yes). | |
| **Test purpose:** | Ensure that when user A calls busy user B, the call is forwarded to user C, user A is notified of call diversion and informed of the diverted-to number (user C has presentation allowed - no COLR) and user C is informed of the forwarding number (user B has presentation allowed). | |
| **Parameter values:** | BC=speech, CF active | |
| **Comments:** | | |

| 211202 | Ref. to ETS 300 207-1 [16], subclauses 9.2.2, 9.2.4.3, 9.2.5 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/CFB/TC211202 | |
| **Selection criteria:** | The user A and the user C are in network N1 and user C is provided with COLR. The user B is in network N2 and is provided with CFB ("calling user is notified of call diversion"=Yes, with diverted-to number, "diverting number is released to the diverted-to user"=No). | |
| **Test purpose:** | Ensure that when user A calls busy user B, the call is forwarded to user C, user A is notified of call diversion and not informed of the diverted-to number (user C has presentation not allowed - COLR) and user C is not informed of the forwarding number (user B has presentation not allowed). | |
| **Parameter values:** | BC=speech, CF active | |
| **Comments:** | | |

| 211301 | Ref. to ETS 300 102-1 [1], subclauses 9.2.2, 9.2.4.4, 9.2.5 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/CFNR/TC211301 | |
| **Selection criteria:** | The user A and the user C are in network N1. The user B is in network N2 and is provided with CFNR ("calling user is notified of call diversion"=Yes, with diverted-to number, "diverting number is released to the diverted-to user"=Yes). | |
| **Test purpose:** | Ensure that when user A calls user B, if unanswered, the call is forwarded to user C, user A is notified of call diversion and informed of the diverted-to number (user C has presentation allowed - no COLR) and user C is informed of the forwarding number (user B has presentation allowed). | |
| **Parameter values:** | BC=speech, CF active | |
| **Comments:** | | |

| 211302 | Ref. to ETS 300 102-1 [1], subclauses 9.2.2, 9.2.4.4, 9.2.5 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/CFNR/TC211302 | |
| **Selection criteria:** | The user A and the user C are in network N1 and user C is provided with COLR. The user B is in network N2 and is provided with CFNR ("calling user is notified of call diversion"=Yes, with diverted-to number, "diverting number is released to the diverted-to user"=No). | |
| **Test purpose:** | Ensure that when user A calls user B, if unanswered the call is forwarded to user C, user A is notified of call diversion and not informed of the diverted-to number (user C has presentation not allowed - COLR) and user C is not informed of the forwarding number (user B has presentation not allowed). | |
| **Parameter values:** | BC=speech, CF active | |
| **Comments:** | | |

| 211401 | Ref. to ETS 300 207-1 [16], subclauses 9.2.2, 9.2.4.5, 9.2.5 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/CD/TC211401 | |
| **Selection criteria:** | The user A and the user C are in network N1. The user B is in network N2 and is provided with CD ("calling user is notified of call diversion"=Yes, with diverted-to number, "diverting number is released to the diverted-to user"=Yes). | |
| **Test purpose:** | Ensure that when user A calls user B and user B invoke CD (with the address of user C) during the alerting phase, the call is deflected to user C, user A is notified of call diversion and informed of the diverted-to number (user C has presentation allowed - no COLR) and user C is informed of the forwarding number (user B has presentation allowed). | |
| **Parameter values:** | BC=speech | |
| **Comments:** | P104 TC21301 | |

| 211402 | Ref. to ETS 300 207-1 [16], subclauses 9.2.2, 9.2.4.5, 9.2.5 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/CD/TC211402 | |
| **Selection criteria:** | The user A and the user C are in network N1. The user B is in network N2 and is provided with CD ("calling user is notified of call diversion"=Yes, with diverted-to number, "diverting number is released to the diverted-to user"=Yes). | |
| **Test purpose:** | Ensure that when user A calls user B and user B invoke CD (with the address of user C) during call present state, the call is deflected to user C, user A is notified of call diversion and informed of the diverted-to number (user C has presentation allowed - no COLR) and user C is informed of the forwarding number (user B has presentation allowed). | |
| **Parameter values:** | BC=speech | |
| **Comments:** | P104 TC21302 | |

| 211501 | Ref. to ETS 300 210-1 [17], subclause 9.2.2.1 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/FPH/TC211501 | |
| **Selection criteria:** | The called (served) user is a freephone subscriber | |
| **Test purpose:** | Ensure that call establishment to a Freephone subscriber is possible and that the calling user is not charged. Verify that the free phone indication is delivered correctly to the called user. | |
| **Parameter values:** | BC=speech | |
| **Comments:** | P104 TC21401 | |

| 211601 | Ref. to ETS 300 130-1 [18], subclause 9.2.1 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/MCID/TC211601 | |
| **Selection criteria:** | The called (served) user is provided with MCID | |
| **Test purpose:** | Ensure that if MCID is invoked by the called user in the Active call state, the call is registered. | |
| **Parameter values:** | BC=speech | |
| **Comments:** | P104 TC21501 | |

| 211602 | Ref. to ETS 300 130-1 [18], subclause 9.2.1 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/MCID/TC211602 | |
| **Selection criteria:** | The called user is provided with MCID | |
| **Test purpose:** | Ensure that if MCID in invoked by the called user in the Disconnect Indication call state, the call is registered. | |
| **Parameter values:** | BC=speech | |
| **Comments:** | P104 TC21502 | |

| 211701 | Ref. to ETS 300 188-1 [19], subclause 9.2 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/3PTY/TC211701 | |
| **Selection criteria:** | The user A is in network N1 and is provided with 3PTY. The user B and user C are in the network N2. | |
| **Test purpose:** | Ensure that user A can establish a three-way conversation call with user B and user C, and the remote parties are notified of the three-way conversation establishment and disconnection. | |
| **Parameter values:** | BC=speech | |
| **Comments:** | P104 TC21601 | |

| 211801 | Ref. to ETS 300 141-1 [20], clause 7 | Other relevant ref.: ETS 300 196-1 [25], subclause 7.1 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/HOLD/TC211801 | |
| **Selection criteria:** | The calling user is provided with HOLD | |
| **Test purpose:** | Ensure that the remote user is notified of the call hold and retrieval | |
| **Parameter values:** | BC=speech | |
| **Comments:** | P104 TC21701 | |

| 211901 | Ref. to ETS 300 058-1 [21], clause 7 | Other relevant ref.: ETS 300 102-1 [1], subclause 4.5.2.1 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/CW/TC211901 | |
| **Selection criteria:** | The called user is provided with CW, notification allowed | |
| **Test purpose:** | Ensure that when all B-channels are busy at the called side, the calling user is notified of the call waiting. | |
| **Parameter values:** | BC=speech | |
| **Comments:** | P104 TC21801 | |

| 212101 | Ref. to ETS 300 369-1 [23], subclauses 9.2.1, 9.2.3, 9.2.4 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/ECT/TC212101 | |
| **Selection criteria:** | The user A is in network N1 and is provided with ECT. The user B and user C are in network N2. | |
| **Test purpose:** | Ensure that when user A invoke ECT (using implicit linkage) with a call to user B held and a call to user C in the Active state, both calls are released and a connection is established between user B and user C. A notification of call transfer is sent to user B and user C with information of the transferred-to number (user B and user C have presentation allowed - no COLR). | |
| **Parameter values:** | BC=speech | |
| **Comments:** | | |

| 212102 | Ref. to ETS 300 369-1 [23], subclauses 9.2.1, 9.2.3, 9.2.4 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/ECT/TC212102 | |
| **Selection criteria:** | The user A is in network N1 and is provided with ECT. The user B and user C are in network N2 and are provided with COLR. | |
| **Test purpose:** | Ensure that when user A invoke ECT (using implicit linkage) with a call to user B held and a call to user C in the Active state, both calls are released and a connection is established between user B and user C. A notification of call transfer is sent to user B and user C without information of the transferred-to number (user B and user C have presentation restricted - COLR). | |
| **Parameter values:** | BC=speech | |
| **Comments:** | | |

| 213001 | Ref. to ETS 300 195-1 [24], clause 5 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/Comb/TC213001 | |
| **Selection criteria:** | The calling user is provided with COLP, UUS1 implicit request and belong to a CUG with outgoing access allowed, the called user is provided with CLIP and SUB | |
| **Test purpose:** | Ensure that when Calling party number is provided by the calling user with Calling party subaddress, Called party subaddress and User-user information elements, all the information elements are correctly delivered to the called user in the SETUP message sent by the network to the called user and when the Connected number is provided by the called user with Connected subaddress and User-user information elements, all the information elements are correctly delivered in the CONNECT message sent by the network to the calling user | |
| **Parameter values:** | BC=speech, HLC=telephony, UI length=32, SI=UPVP | |
| **Comments:** | P104 TC22101* | |

| 213002 | Ref. to ETS 300 195-1 [24], subclauses 5.29, 5.27 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/Comb/TC213002 | |
| **Selection criteria:** | The calling user is provided with COLP, UUS1 implicit request , the called user is provided with CLIP and CFU, the forwarded-to user is provided with CLIP | |
| **Test purpose:** | Ensure that when Calling party number is provided by the calling user with Calling party subaddress and User-user information elements, all the information elements are correctly transferred to the forwarded-to user and delivered in the SETUP message sent by the network to the forwarded-to user without any bearer/teleservice change, and when the Connected number is provided by the forwarded-to user with Connected subaddress and User-user information elements, all the information elements are correctly delivered in the CONNECT message sent by the network to the calling user | |
| **Parameter values:** | BC=speech, HLC=telephony, UI length=32, SI=UPVP | |
| **Comments:** | P104 TC22102* | |

| 213003 | Ref. to ETS 300 195-1 [24], clause 5 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/Speech/Comb/TC213003 | |
| **Selection criteria:** | The called user is Freephone subscriber provided with CLIP | |
| **Test purpose:** | Ensure that when Calling party number is provided by the calling user, the Calling party number information element is delivered correctly to the called user | |
| **Parameter values:** | BC=speech, SI=UPVP | |
| **Comments:** | P104 TC22103 | |

#### 6.2.1.2.2 Bearer capability "unrestricted digital information"

| 220101 | Ref. to ETS 300 092-1 [7], subclause 9.3 | Other relevant ref.: ETS 300 102-1 [1], subclauses 4.5.10, 4.5.11 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/CLIP/TC220101 | |
| **Selection criteria:** | The called user is provided with CLIP | |
| **Test purpose:** | Ensure that when Calling party number is provided by the calling user, Type of number "subscriber number", with Calling party subaddress, the Calling party number and Calling party subaddress information elements are correctly delivered to the called (served) user. | |
| **Parameter values:** | BC=UDI, SI=UPVP, TON=international | |
| **Comments:** | | |

| 220102 | Ref. to ETS 300 092-1 [7], subclause 9.3 | Other relevant ref.: ETS 300 102-1 [1], subclauses 4.5.10, 4.5.11 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/CLIP/TC220102 | |
| **Selection criteria:** | The called user is provided with CLIP | |
| **Test purpose:** | Ensure that when Calling party number is provided by the calling user, Type of number "national number", with Calling party subaddress, the Calling party number and Calling party subaddress information elements are correctly delivered to the called (served) user. | |
| **Parameter values:** | BC=UDI, SI=UPVP, TON=international | |
| **Comments:** | | |

| 220103 | Ref. to ETS 300 092-1 [7], subclause 9.3 | Other relevant ref.: ETS 300 102-1 [1], subclauses 4.5.10, 4.5.11 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/CLIP/TC220103 | |
| **Selection criteria:** | The called user is provided with CLIP | |
| **Test purpose:** | Ensure that when Calling party number is provided by the calling user, Type of number "international number", with Calling party subaddress, the Calling party number and Calling party subaddress information elements are correctly delivered to the called (served) user. | |
| **Parameter values:** | BC=UDI, SI=UPVP, TON=international | |
| **Comments:** | | |

| 220104 | Ref. to ETS 300 092-1 [7], subclause 9.3 | Other relevant ref.: ETS 300 102-1 [1], subclauses 4.5.10, 4.5.11 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/CLIP/TC220104 | |
| **Selection criteria:** | The called user is provided with CLIP | |
| **Test purpose:** | Ensure that when Calling party number is provided by the calling user, Type of number "unknown", with Calling party subaddress, the Calling party number and Calling party subaddress information elements are correctly delivered to the called (served) user. | |
| **Parameter values:** | BC=UDI, SI=UPVP, TON=international | |
| **Comments:** | | |

| 220105 | Ref. to ETS 300 092-1 [7], subclause 9.3 | Other relevant ref.: ETS 300 102-1 [1], subclause 4.5.10 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/CLIP/TC220105 | |
| **Selection criteria:** | The called user is provided with CLIP | |
| **Test purpose:** | Ensure that when no Calling party number information element is provided by the calling user, (and no Calling party subaddress), the Calling party number information element is network provided and correctly delivered to the called (served) user. | |
| **Parameter values:** | BC=UDI, SI=NP, TON=international | |
| **Comments:** | | |

| 220201 | Ref. to ETS 300 093-1 [8], subclause 9.4.1 | Other relevant ref.: ETS 300 092-1 [7], subclause 9.5.1 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/CLIR/TC220201 | |
| **Selection criteria:** | The calling user is provided with CLIR permanent mode subscription, the called user with CLIP | |
| **Test purpose:** | Ensure that when the Calling party number is provided by the calling user, with Calling party subaddress, the Calling party number information element is delivered to the called party without any digit information. The Calling party subaddress shall not be present | |
| **Parameter values:** | BC=UDI, PI=PR, SI=NP, TON=unknown, NPI=unknown | |
| **Comments:** | | |

| 220202 | Ref. to ETS 300 093-1 [8], subclause 9.4.1 | Other relevant ref.: ETS 300 092-1 [7], subclause 9.5.1 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/CLIR/TC220202 | |
| **Selection criteria:** | The calling user is provided with CLIR permanent mode subscription, the called user with CLIP | |
| **Test purpose:** | Ensure that when no Calling party number is provided by the calling user (and no Calling party subaddress), the Calling party number information element is network provided and delivered to the called user without any digit information. | |
| **Parameter values:** | BC=UDI, PI=PR, SI=NP, TON=unknown, NPI=unknown | |
| **Comments:** | | |

| 220301 | Ref. to ETS 300 097-1 [9], subclause 9.5.1 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/COLP/TC220301 | |
| **Selection criteria:** | The calling user is provided with COLP | |
| **Test purpose:** | Ensure that when the Connected number is provided by the called user, Type of number "subscriber number", with Connected subaddress, the Connected number and Connected subaddress information elements are correctly delivered to the calling (served) user. | |
| **Parameter values:** | BC=UDI, SI=UPVP, TON=international | |
| **Comments:** | | |

| 220302 | Ref. to ETS 300 097-1 [9], subclause 9.5.1 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/COLP/TC220302 | |
| **Selection criteria:** | The calling user is provided with COLP | |
| **Test purpose:** | Ensure that when the Connected number is provided by the called user, Type of number "national number", with Connected subaddress, the Connected number and Connected subaddress information elements are correctly delivered to the calling (served) user. | |
| **Parameter values:** | BC=UDI, SI=UPVP, TON=international | |
| **Comments:** | | |

| 220303 | Ref. to ETS 300 097-1 [9], subclause 9.5.1 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/COLP/TC220303 | |
| **Selection criteria:** | The calling user is provided with COLP | |
| **Test purpose:** | Ensure that when the Connected number is provided by the called user, Type of number "international number", with Connected subaddress, the Connected number and Connected subaddress information elements are correctly delivered to the calling (served) user. | |
| **Parameter values:** | BC=UDI, SI=UPVP, TON=international | |
| **Comments:** | | |

| 220304 | Ref. to ETS 300 097-1 [9], subclause 9.5.1 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/COLP/TC220304 | |
| **Selection criteria:** | The calling user is provided with COLP | |
| **Test purpose:** | Ensure that when the Connected number is provided by the called user, Type of number "unknown", with Connected subaddress, the Connected number and Connected subaddress information elements are correctly delivered to the calling (served) user. | |
| **Parameter values:** | BC=UDI, SI=UPVP, TON=international | |
| **Comments:** | | |

| 220305 | Ref. to ETS 300 097-1 [9], subclause 9.5.1 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/COLP/TC220305 | |
| **Selection criteria:** | Calling user is provided with COLP | |
| **Test purpose:** | Ensure that when no Connected number is provided by the called user (and no Connected subaddress), the Connected number information element is network provided and correctly delivered to the calling (served) user. | |
| **Parameter values:** | BC=UDI, SI=NP, TON=international | |
| **Comments:** | | |

| 220401 | Ref. to ETS 300 098-1 [10], subclauses 9.3.1, 9.4.1 | Other relevant ref.: ETS 300 097-1 [9], subclause 9.5.1 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/COLR/TC220401 | |
| **Selection criteria:** | The called (served) user is provided with COLR permanent mode subscription, the calling user with COLP | |
| **Test purpose:** | Ensure that when the Connected number is provided by the called user, with Connected subaddress, the Connected number information element is delivered to the calling user without any digit information. The Connected subaddress shall not be present | |
| **Parameter values:** | BC=UDI, PI=PR, SI=NP, TON=unknown, NPI=unknown | |
| **Comments:** | | |

| 220402 | Ref. to ETS 300 098-1 [10], subclauses 9.3.1, 9.4.1 | Other relevant ref.: ETS 300 097-1 [9], subclause 9.5.1 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/COLR/TC220402 | |
| **Selection criteria:** | The called (served) user is provided with COLR permanent mode subscription, the calling user with COLP | |
| **Test purpose:** | Ensure that when no Connected number is provided by the called user (and no Connected subaddress), the Connected number information element is network provided and delivered to the Calling party without any digit information. | |
| **Parameter values:** | BC=UDI, PI=PR, SI=NP, TON=unknown, NPI=unknown | |
| **Comments:** | | |

| 220501 | Ref. to ETS 300 138-1 [11], subclauses 9.2.2, 9.2.4 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/CUG/TC220501 | |
| **Selection criteria:** | Calling user and called user belong to the same (international) CUG | |
| **Test purpose:** | Ensure that when the calling user and called user belongs to the same (international) CUG, call establishment is possible | |
| **Parameter values:** | BC=UDI, CUG default request | |
| **Comments:** | | |

| 220502 | Ref. to ETS 300 138-1 [11], subclauses 9.2.2, 9.2.4 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/CUG/TC220502 | |
| **Selection criteria:** | Calling user belongs to a CUG with outgoing access allowed and called user belongs to another CUG with incoming access allowed | |
| **Test purpose:** | Ensure that when the calling user belongs to a CUG with outgoing access allowed and the called user belongs to another CUG with incoming access allowed, the call establishment is possible | |
| **Parameter values:** | BC=UDI, CUG default request | |
| **Comments:** | | |

| 220503 | Ref. to ETS 300 138-1 [11], subclause 9.2.2 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/CUG/TC220503 | |
| **Selection criteria:** | Calling user belongs to a CUG with outgoing access "allowed" and the called user is not a CUG subscriber | |
| **Test purpose:** | Ensure that when the calling user belongs to a CUG with outgoing access allowed and the called user is not a CUG subscriber, the call establishment is possible | |
| **Parameter values:** | BC=UDI, CUG default request | |
| **Comments:** | | |

| 220504 | Ref. to ETS 300 138-1 [11], subclause 9.2.3 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/CUG/TC220504 | |
| **Selection criteria:** | Called user belong to a CUG with incoming access "not allowed" and calling user is not member of CUG | |
| **Test purpose:** | Ensure that when the called user belongs to a CUG with incoming access "not allowed", call establishment is not possible and the network initiate call clearing to the calling user with cause value #87 "user not a member of CUG". | |
| **Parameter values:** | BC=UDI | |
| **Comments:** | | |

| 220505 | Ref. to ETS 300 138-1 [11], subclause 9.2.2 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/CUG/TC220505 | |
| **Selection criteria:** | Calling user belong to a CUG with outgoing access "not allowed" and called user is not member of CUG | |
| **Test purpose:** | Ensure that when the calling user belong to CUG with outgoing access "not allowed", call establishment is not possible and the network initiate call clearing to the calling user with cause value #87 "user not a member of CUG". | |
| **Parameter values:** | BC=UDI, CUG default request | |
| **Comments:** | | |

| 220601 | Ref. to ETS 300 061-1 [12], subclause 9.2 | Other relevant ref.: ETS 300 102-1 [1], subclause 4.5.9 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/SUB/TC220601 | |
| **Selection criteria:** | The called (served) user is provided with SUB | |
| **Test purpose:** | Ensure that when the Called party subaddress is provided by the calling user, the Called party subaddress is correctly delivered to the called (served) user | |
| **Parameter values:** | BC=UDI | |
| **Comments:** | | |

| 220602 | Ref. to ETS 300 061-1 [12], 9.2 | Other relevant ref.: ETS 300 102-1 [1], 4.5.9 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/SUB/TC220602 | |
| **Selection criteria:** | The called (served) user is provided with SUB | |
| **Test purpose:** | Ensure that when the Called party subaddress is provided by the calling user with length=minimum, the Called party subaddress is correctly delivered to the called (served) user without any digit information | |
| **Parameter values:** | BC=UDI | |
| **Comments:** | | |

| 220701 | Ref. to ETS 300 055-1 [13], subclause 9.2.1 | Other relevant ref.: ETS 300 102-1 [1], subclause 5.6 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/TP/TC220701 | |
| **Selection criteria:** | The calling user shall be a basic access | |
| **Test purpose:** | Ensure that the called user is notified of the call suspension and resumption by the calling user (no call identity is used) | |
| **Parameter values:** | BC=UDI | |
| **Comments:** | | |

| 220702 | Ref. to ETS 300 055-1 [13], subclause 9.2.1 | Other relevant ref.: ETS 300 102-1 [1], subclause 5.6 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/TP/TC220702 | |
| **Selection criteria:** | The called user shall be a basic access | |
| **Test purpose:** | Ensure that the calling user is notified of the call suspension and resumption by the called user (no call identity is used) | |
| **Parameter values:** | BC=UDI | |
| **Comments:** | | |

| 220703 | Ref. to ETS 300 055-1 [13], subclause 9.2.2 | Other relevant ref.: ETS 300 102-1 [1], subclause 5.6.5 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/TP/TC220703 | |
| **Selection criteria:** | The calling user shall be a basic access | |
| **Test purpose:** | Ensure that when the call is suspended, with the expiry of timer T307 before the call re-establishment, the network starts call clearing to the (still) active side with cause value #102 "recovery on timer expiry". | |
| **Parameter values:** | BC=UDI | |
| **Comments:** | | |

| 220801 | Ref. to ETS 300 286-1 [14], subclauses 9.1.1.1, 9.1.2.1 | Other relevant ref.: ETS 300 102-1 [1], subclause 4.5.29 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/UUS1/TC220801 | |
| **Selection criteria:** | The calling (served) user is provided with UUS1 implicit request | |
| **Test purpose:** | Ensure that the network can transport a User-user information element included in the SETUP message sent from the calling user and delivered in the SETUP message sent by the network to the called user | |
| **Parameter values:** | BC=UDI, UI length=32 | |
| **Comments:** | | |

| 220802 | Ref. to ETS 300 286-1 [14], subclause 9.1.2.1 | Other relevant ref.: ETS 300 102-1 [1], subclause 4.5.29 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/UUS1/TC220802 | |
| **Selection criteria:** | The calling (served) user is provided with UUS1 implicit request | |
| **Test purpose:** | Ensure that after implicit activation of UUS1, the network can transport a User-user information element included in the ALERTING message sent from the called user to the calling user. | |
| **Parameter values:** | BC=UDI, UI length=32 | |
| **Comments:** | | |

| 220803 | Ref. to ETS 300 286-1 [14], subclause 9.1.2.1 | Other relevant ref.: ETS 300 102-1 [1], subclause 4.5.29 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/UUS1/TC220803 | |
| **Selection criteria:** | The calling (served) user is provided with UUS1 implicit request | |
| **Test purpose:** | Ensure that after implicit activation of UUS1, the network can transport a User-user information element included in the CONNECT message sent from the called user to the calling user | |
| **Parameter values:** | BC=UDI, UI length=32 | |
| **Comments:** | | |

| 220804 | Ref. to ETS 300 286-1 [14], subclause 9.1.2.2.1a | Other relevant ref.: ETS 300 102-1 [1] |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/UUS1/TC220804 | |
| **Selection criteria:** | The calling (served) user is provided with UUS1 implicit request | |
| **Test purpose:** | Ensure that after implicit activation of UUS1 and with the call in the active state, the network can transport a User-user information element included in a call clearing DISCONNECT message sent from the calling user and delivered in the DISCONNECT message sent by the network to the called user. | |
| **Parameter values:** | BC=UDI, UI length=32 | |
| **Comments:** | | |

| 220805 | Ref. to ETS 300 286-1 [14], subclause 9.1.2.2.1b | Other relevant ref.: ETS 300 102-1 [1] |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/UUS1/TC220805 | |
| **Selection criteria:** | The calling (served) user is provided with UUS1 implicit request. | |
| **Test purpose:** | Ensure that after implicit activation of UUS1, the network can transport a User-user information element included in premature clearing RELEASE COMPLETE message sent from the called user and delivered in the DISCONNECT message sent by the network to the calling user | |
| **Parameter values:** | BC=UDI, UI length=32 | |
| **Comments:** | | |

| 220806 | Ref. to ETS 300 286-1 [14], subclause 9.1.1.1.1 | Other relevant ref.: ETS 300 102-1 [1], subclause 7.1.3.2 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/UUS1/TC220806 | |
| **Selection criteria:** | The calling (served) user is provided with UUS1 implicit request | |
| **Test purpose:** | Ensure that implicit activation of UUS1 with a User-user information element with the minimum length of three octets (without any user information), included in the SETUP message sent from the calling user, is supported | |
| **Parameter values:** | BC=UDI | |
| **Comments:** | | |

| 220807 | Ref. to ETS 300 286-1 [14], subclause 9.1.2.2.1b | Other relevant ref.: ETS 300 102-1 [1] |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/UUS1/TC220807 | |
| **Selection criteria:** | The calling (served) user is provided with UUS1 implicit request | |
| **Test purpose:** | Ensure that after implicit activation of UUS1 and with the call in the active state, the network can transport a User-user information element included in a call clearing DISCONNECT message sent from the called user and delivered in the DISCONNECT message sent by the network to the calling user. | |
| **Parameter values:** | BC=UDI, UI length=32 | |
| **Comments:** | | |

| 220808 | Ref. to ETS 300 286-1 [14], subclause 9.1.2.2.1b | Other relevant ref.: ETS 300 102-1 [1] |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/UUS1/TC220808 | |
| **Selection criteria:** | The calling (served) user is provided with UUS1 implicit request<br>Multipoint configuration for the called side. | |
| **Test purpose:** | Ensure that after implicit activation of UUS1, the network can transport a User-user information element associated with the highest priority cause included in premature clearing RELEASE COMPLETE messages sent from the called users and delivered in the DISCONNECT message sent by the network to the calling user | |
| **Parameter values:** | BC=UDI, UI length=32 | |
| **Comments:** | | |

| 220809 | Ref. to ETS 300 286-1 [14], subclause 9.1.2.2.1b | Other relevant ref.: ETS 300 102-1 [1] |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/UUS1/TC220809 | |
| **Selection criteria:** | The calling (served) user is provided with UUS1 implicit request. | |
| **Test purpose:** | Ensure that after implicit activation of UUS1, the network can transport a User-user information element in premature clearing DISCONNECT message sent from the called user and delivered in the DISCONNECT message sent by the network to the calling user | |
| **Parameter values:** | BC=UDI, UI length=32 | |
| **Comments:** | | |

| 220810 | Ref. to ETS 300 286-1 [14], subclause 9.1.2.1 | Other relevant ref.: ETS 300 102-1 [1], subclause 4.5.29 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/UUS1/TC220810 | |
| **Selection criteria:** | The calling (served) user is provided with UUS1 implicit request.<br>Multipoint configuration for the called side. | |
| **Test purpose:** | Ensure that after implicit activation of UUS1, the network can transport a User-user information element included in the first received ALERTING message sent from the called users to the calling user in a ALERTING message | |
| **Parameter values:** | BC=UDI, UI length=32 | |
| **Comments:** | | |

| 220811 | Ref. to ETS 300 286-1 [14], subclause 9.1.2.2.1b | Other relevant ref.: ETS 300 102-1 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/UUS1/TC220811 | |
| **Selection criteria:** | The calling (served) user is provided with UUS1 implicit request.<br>Multipoint configuration for the called side. | |
| **Test purpose:** | Ensure that after implicit activation of UUS1, the network can transport a User-user information element associated with the highest priority cause included in premature clearing DISCONNECT messages sent from the called users and delivered in the DISCONNECT message sent by the network to the calling user | |
| **Parameter values:** | BC=UDI, UI length=32 | |
| **Comments:** | | |

| 220831 | Ref. to ETS 300 286-1 [14], subclause 9.1.1.2.2 | Other relevant ref.: ETS 300 102-1 [1], subclause 7.1.3.6 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/UUS1/TC220831 | |
| **Selection criteria:** | The calling (served) user is provided with UUS1 explicit request | |
| **Test purpose:** | Ensure that after explicit request of UUS1 indicating "required", if the called user rejects the call with a RELEASE COMPLETE message indicating cause value #29 "facility rejected", the network transport the cause value to the calling user. A UUS1 rejection with Error value "rejectedByUser" shall be included in the message | |
| **Parameter values:** | BC=UDI | |
| **Comments:** | | |

| 220832 | Ref. to ETS 300 286-1 [14], subclause 9.1.1.2.2 | Other relevant ref.: ETS 300 102-1 [1], subclause 7.1.3.6 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/UUS1/TC220832 | |
| **Selection criteria:** | The calling (served) user is provided with UUS1 explicit request | |
| **Test purpose:** | Ensure that after explicit request of UUS1 indicating "required", if the destination network rejects the UUS1 request, the network initiates call clearing to the calling user with cause value #69 "requested facility not implemented" or cause value #29 "facility rejected". A UUS1 rejection with the Error value "rejectedByUser" shall be included in the message. | |
| **Parameter values:** | BC=UDI | |
| **Comments:** | | |

| 220833 | Ref. to ETS 300 286-1 [14], subclause 9.1.1.2.1 | Other relevant ref.: ETS 300 102-1 [1], subclause 7.1.3.2 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/UUS1/TC220833 | |
| **Selection criteria:** | The calling (served) user is provided with UUS1 explicit request | |
| **Test purpose:** | Ensure that with the explicit request of UUS1 indicating "preferred", the network can transport a User-user information element included in the SETUP message sent from the calling user and delivered in the SETUP message sent by the network to the called user, and return the explicit service acceptance from the called user | |
| **Parameter values:** | BC=UDI, UI length=32 | |
| **Comments:** | | |

| 220851 | Ref. to ETS 300 286-1 [14], subclause 9.2.2.1 | Other relevant ref.: ETS 300 102-1 [1], subclause 7.1.4.4 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/UUS2/TC220851 | |
| **Selection criteria:** | The calling (served) user is provided with UUS2.<br>Point-to-point configuration for the called side. | |
| **Test purpose:** | Ensure that after activation of UUS2 indicating "preferred", the network can transport two USER INFORMATION messages in each direction, between the ALERTING and the CONNECT messages. | |
| **Parameter values:** | BC=UDI, UI length=32 | |
| **Comments:** | | |

| 220852 | Ref. to ETS 300 286-1 [14], subclause 9.2.1.2 | Other relevant ref.: ETS 300 102-1 [1], subclause 7.1.4.3 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/UUS2/TC220852 | |
| **Selection criteria:** | The calling (served) user is provided with UUS2.<br>Point-to-point configuration for the called side. | |
| **Test purpose:** | Ensure that after activation of UUS2 indicating "preferred", if the network does not receive an explicit acceptance or rejection in the ALERTING message from the called user, a UUS2 rejection with the Error value "rejected by the user, it is returned to the calling user in a ALERTING message sent from the network and the call can be established. | |
| **Parameter values:** | BC=UDI | |
| **Comments:** | | |

| 220871 | Ref. to ETS 300 286-1 [14], subclause 9.3.1.1 | Other relevant ref.: ETS 300 102-1 [1], subclause 7.1.5.2 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/UUS3/TC220871 | |
| **Selection criteria:** | The calling (served) user is provided with UUS3 | |
| **Test purpose:** | Ensure that after activation of UUS3 during call establishment indicating "preferred", the network can transport USER INFORMATION messages in both directions during the Active state of the call | |
| **Parameter values:** | BC=UDI, UI length=32 | |
| **Comments:** | | |

| 220872 | Ref. to ETS 300 286-1 [14], subclause 9.3.1.1.2 | Other relevant ref.: ETS 300 102-1 [1], subclause 7.1.5.3 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/UUS3/TC220872 | |
| **Selection criteria:** | The calling (served) user is provided with UUS3 | |
| **Test purpose:** | Ensure that after the calling user request UUS3 during call establishment indicating "preferred", if the network does not receive an explicit acceptance or rejection in the CONNECT message from the called user, a UUS3 rejection with the Error value "rejected by the user" is included in the CONNECT message sent to the calling user. | |
| **Parameter values:** | BC=UDI | |
| **Comments:** | | |

| 220873 | Ref. to ETS 300 286-1 [14], subclause 9.3.1.1 | Other relevant ref.: ETS 300 102-1 [1], subclause 7.1.5.2 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/UUS3/TC220873 | |
| **Selection criteria:** | The calling (served) user is provided with UUS3 | |
| **Test purpose:** | Ensure that after activation of UUS3 during call establishment indicating "required", the network can transport USER INFORMATION messages in both directions during the Active state of the call | |
| **Parameter values:** | BC=UDI, UI length=32 | |
| **Comments:** | | |

| 220874 | Ref. to ETS 300 286-1 [14], subclause 9.3.1.2.1 | Other relevant ref.: ETS 300 102-1 [1], subclause 7.1.5.4 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/UUS3/TC220874 | |
| **Selection criteria:** | The calling (served) user is provided with UUS3 | |
| **Test purpose:** | Ensure that after activation of UUS3 during the Active call state indicating "preferred", the network can transport USER INFORMATION messages in both directions during the Active state of the call | |
| **Parameter values:** | BC=UDI, UI length=32 | |
| **Comments:** | | |

| 220875 | Ref. to ETS 300 286-1 [14], subclause 9.3.1.2.2 | Other relevant ref.: ETS 300 102-1 [1], subclause 7.1.5.5 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/UUS3/TC220875 | |
| **Selection criteria:** | The calling (served) user is provided with UUS3 | |
| **Test purpose:** | Ensure that after the calling user request UUS3 during the Active call state indicating "preferred", if the called user rejects the service 3 request, the network can transport the FACILITY message including a UUS3 rejection with the Error value "rejected by the user" from the called user to the calling user. | |
| **Parameter values:** | BC=UDI | |
| **Comments:** | | |

| 221101 | Ref. to ETS 300 207-1 [16], subclauses 9.2.2, 9.2.5 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/CFU/TC221101 | |
| **Selection criteria:** | The user A and the user C are in network N1. The user B is in network N2 and is provided with CFU ("calling user is notified of call diversion"=Yes, with diverted-to number, "diverting number is released to the diverted-to user"=Yes). | |
| **Test purpose:** | Ensure that when user A calls user B, the call is forwarded to user C, user A is notified of call diversion and informed of the diverted-to number (user C has presentation allowed - no COLR) and user C is informed of the forwarding number (user B has presentation allowed). | |
| **Parameter values:** | BC=UDI, CF active | |
| **Comments:** | | |

| 221102 | Ref. to ETS 300 207-1 [16], subclauses 9.2.2, 9.2.5 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/CFU/TC221102 | |
| **Selection criteria:** | The user A and the user C are in network N1 and user C is provided with COLR. The user B is in network N2 and is provided with CFU ("calling user is notified of call diversion"=Yes, with diverted-to number, "diverting number is released to the diverted-to user"=No). | |
| **Test purpose:** | Ensure that when user A calls user B, the call is forwarded to user C, user A is notified of call diversion and not informed of the diverted-to number (user C has presentation not allowed - COLR) and user C is not informed of the forwarding number (user B has presentation not allowed). | |
| **Parameter values:** | BC=UDI, CF active | |
| **Comments:** | | |

| 221201 | Ref. to ETS 300 207-1 [16], subclauses 9.2.2, 9.2.4.3, 9.2.5 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/CFB/TC221201 | |
| **Selection criteria:** | The user A and the user C are in network N1. The user B is in network N2 and is provided with CFB ("calling user is notified of call diversion"=Yes, with diverted-to number, "diverting number is released to the diverted-to user"=Yes). | |
| **Test purpose:** | Ensure that when user A calls busy user B, the call is forwarded to user C, user A is notified of call diversion and informed of the diverted-to number (user C has presentation allowed - no COLR) and user C is informed of the forwarding number (user B has presentation allowed). | |
| **Parameter values:** | BC=UDI, CF active | |
| **Comments:** | | |

| 221202 | Ref. to ETS 300 207-1 [16], subclauses 9.2.2, 9.2.4.3, 9.2.5 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/CFB/TC221202 | |
| **Selection criteria:** | The user A and the user C are in network N1 and user C is provided with COLR. The user B is in network N2 and is provided with CFB ("calling user is notified of call diversion"=Yes, with diverted-to number, "diverting number is released to the diverted-to user"=No). | |
| **Test purpose:** | Ensure that when user A calls busy user B, the call is forwarded to user C, user A is notified of call diversion and not informed of the diverted-to number (user C has presentation not allowed - COLR) and user C is not informed of the forwarding number (user B has presentation not allowed). | |
| **Parameter values:** | BC=UDI, CF active | |
| **Comments:** | | |

| 221301 | Ref. to ETS 300 102-1 [1], subclauses 9.2.2, 9.2.4.4, 9.2.5 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/CFNR/TC221301 | |
| **Selection criteria:** | The user A and the user C are in network N1. The user B is in network N2 and is provided with CFNR ("calling user is notified of call diversion"=Yes, with diverted-to number, "diverting number is released to the diverted-to user"=Yes). | |
| **Test purpose:** | Ensure that when user A calls user B, if unanswered the call is forwarded to user C, user A is notified of call diversion and informed of the diverted-to number (user C has presentation allowed - no COLR) and user C is informed of the forwarding number (user B has presentation allowed). | |
| **Parameter values:** | BC=UDI, CF active | |
| **Comments:** | | |

| 221302 | Ref. to ETS 300 102-1 [1], subclauses 9.2.2, 9.2.4.4, 9.2.5 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/CFNR/TC221302 | |
| **Selection criteria:** | The user A and the user C are in network N1 and user C is provided with COLR. The user B is in network N2 and is provided with CFNR ("calling user is notified of call diversion"=Yes, with diverted-to number, "diverting number is released to the diverted-to user"=No). | |
| **Test purpose:** | Ensure that when user A calls user B, if unanswered the call is forwarded to user C, user A is notified of call diversion and not informed of the diverted-to number (user C has presentation not allowed - COLR) and user C is not informed of the forwarding number (user B has presentation not allowed). | |
| **Parameter values:** | BC=UDI, CF active | |
| **Comments:** | | |

| 221401 | Ref. to ETS 300 207-1 [16], subclauses 9.2.2, 9.2.4.5, 9.2.5 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/CD/TC221401 | |
| **Selection criteria:** | The user A and the user C are in network N1. The user B is in network N2 and is provided with CD ("calling user is notified of call diversion"=Yes, with diverted-to number, "diverting number is released to the diverted-to user"=Yes). | |
| **Test purpose:** | Ensure that when user A calls user B and user B invoke CD (with the address of user C) during the alerting phase, the call is deflected to user C, user A is notified of call diversion and informed of the diverted-to number (user C has presentation allowed - no COLR) and user C is informed of the forwarding number (user B has presentation allowed). | |
| **Parameter values:** | BC=UDI | |
| **Comments:** | | |

| 221402 | Ref. to ETS 300 207-1 [16], subclauses 9.2.2, 9.2.4.5, 9.2.5 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/CD/TC221402 | |
| **Selection criteria:** | The user A and the user C are in network N1. The user B is in network N2 and is provided with CD ("calling user is notified of call diversion"=Yes, with diverted-to number, "diverting number is released to the diverted-to user"=Yes). | |
| **Test purpose:** | Ensure that when user A calls user B and user B invoke CD (with the address of user C) during call present state, the call is deflected to user C, user A is notified of call diversion and informed of the diverted-to number (user C has presentation allowed - no COLR) and user C is informed of the forwarding number (user B has presentation allowed). | |
| **Parameter values:** | BC=UDI | |
| **Comments:** | | |

| 221501 | Ref. to ETS 300 210-1 [17], subclause 9.2.2.1 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/FPH/TC221501 | |
| **Selection criteria:** | The called (served) user is a freephone subscriber | |
| **Test purpose:** | Ensure that call establishment to a Freephone subscriber is possible and that the calling user is not charged. Verify that the free phone indication is delivered correctly to the called user. | |
| **Parameter values:** | BC=UDI | |
| **Comments:** | | |

| 221601 | Ref. to ETS 300 130-1 [18], subclause 9.2.1 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/MCID/TC221601 | |
| **Selection criteria:** | The called (served) user is provided with MCID | |
| **Test purpose:** | Ensure that if MCID is invoked by the called user in the Active call state, the call is registered. | |
| **Parameter values:** | BC=UDI | |
| **Comments:** | | |

| 221602 | Ref. to ETS 300 130-1 [18], subclause 9.2.1 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/MCID/TC221602 | |
| **Selection criteria:** | The called user is provided with MCID | |
| **Test purpose:** | Ensure that if MCID in invoked by the called user in the Disconnect Indication call state, the call is registered. | |
| **Parameter values:** | BC=UDI | |
| **Comments:** | | |

| 221801 | Ref. to ETS 300 141-1 [20], clause 7 | Other relevant ref.: ETS 300 196-1 [25], subclause 7.1 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/HOLD/TC221801 | |
| **Selection criteria:** | The calling user is provided with HOLD | |
| **Test purpose:** | Ensure that the remote user is notified of the call hold and retrieval | |
| **Parameter values:** | BC=UDI | |
| **Comments:** | | |

| 221901 | Ref. to ETS 300 058-1 [21], clause 7 | Other relevant ref.: ETS 300 102-1 [1], subclause 4.5.2.1 |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/CW/TC221901 | |
| **Selection criteria:** | The called user is provided with CW, notification allowed | |
| **Test purpose:** | Ensure that when all B-channels are busy at the called side, the calling user is notified of the call waiting. | |
| **Parameter values:** | BC=UDI | |
| **Comments:** | | |

| 222101 | Ref. to ETS 300 369-1 [23], subclauses 9.2.1, 9.2.3, 9.2.4 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/ECT/TC222101 | |
| **Selection criteria:** | The user A is in network N1 and is provided with ECT. The user B and user C are in network N2. | |
| **Test purpose:** | Ensure that when user A invoke ECT (using implicit linkage) with a call to user B held and a call to user C in the Active state, both calls are released and a connection is established between user B and user C. A notification of call transfer is sent to user B and user C with information of the transferred-to number (user B and user C have presentation allowed - no COLR). | |
| **Parameter values:** | BC=UDI | |
| **Comments:** | | |

| 222102 | Ref. to ETS 300 369-1 [23], subclauses 9.2.1, 9.2.3, 9.2.4 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/ECT/TC222102 | |
| **Selection criteria:** | The user A is in network N1 and is provided with ECT. The user B and user C are in network N2 and are provided with COLR. | |
| **Test purpose:** | Ensure that when user A invoke ECT (using implicit linkage) with a call to user B held and a call to user C in the Active state, both calls are released and a connection is established between user B and user C. A notification of call transfer is sent to user B and user C without information of the transferred-to number (user B and user C have presentation restricted - COLR). | |
| **Parameter values:** | BC=UDI | |
| **Comments:** | | |

| 223001 | Ref. to ETS 300 195-1 [24], clause 5 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/Comb/TC223001 | |
| **Selection criteria:** | The calling user is provided with COLP, UUS1 implicit request and belong to a CUG with outgoing access allowed, the called user is provided with CLIP and SUB | |
| **Test purpose:** | Ensure that when Calling party number is provided by the calling user with Calling party subaddress, Called party subaddress and User-user information elements, all the information elements are correctly delivered to the called user in the SETUP message sent by the network to the called user and when the Connected number is provided by the called user with Connected subaddress and User-user information elements, all the information elements are correctly delivered in the CONNECT message sent by the network to the calling user | |
| **Parameter values:** | BC=UDI, HLC=facsimile group 4, LLC=telematic_term,<br>UI length=32, SI=UPVP | |
| **Comments:** | | |

| 223002 | Ref. to ETS 300 195-1 [24], subclauses 5.29, 5.27 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/Comb/TC223002 | |
| **Selection criteria:** | The calling user is provided with COLP, UUS1 implicit request , the called user is provided with CLIP and CFU, the forwarded-to user is provided with CLIP | |
| **Test purpose:** | Ensure that when Calling party number is provided by the calling user with Calling party subaddress and User-user information elements, all the information elements are correctly transferred to the forwarded-to user and delivered in the SETUP message sent by the network to the forwarded-to user without any bearer/teleservice change, and when the Connected number is provided by the forwarded-to user with Connected subaddress and User-user information elements, all the information elements are correctly delivered in the CONNECT message sent by the network to the calling user | |
| **Parameter values:** | BC=UDI, HLC=facsimile group 4, LLC=telematic_term,<br>UI length=32, SI=UPVP | |
| **Comments:** | | |

| 223003 | Ref. to ETS 300 195-1 [24], clause 5 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/Supplementary_services/UDI/Comb/TC223003 | |
| **Selection criteria:** | The called user is Freephone subscriber provided with CLIP | |
| **Test purpose:** | Ensure that when Calling party number is provided by the calling user, Type of number "unknown", the Calling party number information element is delivered correctly to the called user | |
| **Parameter values:** | BC=UDI, SI=UPVP | |
| **Comments:** | | |

#### 6.2.1.3 Test purposes for ISDN-ISDN, B-channel end-to-end performance

| 300101 | Ref. to ETS 300 102-1 [1] | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/B-channel/Speech/TC300101 | |
| **Selection criteria:** | | |
| **Test purpose:** | To ensure that speech transfer on the B-channel is performed correctly. | |
| **Parameter values:** | BC=speech | |
| **Comments:** | | |

| 300201 | Ref. to ETS 300 102-1 [1] | Other relevant ref.: ETS 300 289 [26] |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/B-channel/UDI/TC300201 | |
| **Selection criteria:** | | |
| **Test purpose:** | To verify compliance with the requirements for error and octet slip for the first or the last continuous 24 hour period:<br>- the number of errored seconds shall be less than 5 324;<br>- the number of severely errored seconds shall be less than 105;<br>- the number of octet slips shall be less than 5. | |
| **Parameter values:** | BC=UDI, PRBS=$2^{11}$-1 | |
| **Comments:** | Each direction shall be tested separately | |

| 300301 | Ref. to ETS 300 102-1 [1] | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/B-channel/Audio/TC300301 | |
| **Selection criteria:** | | |
| **Test purpose:** | To ensure that 3,1 kHz signal transfer on the B-channel is performed correctly. | |
| **Parameter values:** | BC=3,1 kHz audio | |
| **Comments:** | | |

| 300401 | Ref. to ETS 300 102-1 [1] | Other relevant ref.: ETS 300 289 [26] |
|---|---|---|
| **TSS reference:** | ISDN-ISDN/B-channel/UDI-TA/TC300401 | |
| **Selection criteria:** | | |
| **Test purpose:** | To verify compliance with the requirements for error and octet slip for the first or the last continuous 24 hour period:<br>- the number of errored seconds shall be less than 5 324;<br>- the number of severely errored seconds shall be less than 105;<br>- the number of octet slips shall be less than 5. | |
| **Parameter values:** | BC=UDI/TA, PRBS=$2^{11}$-1 | |
| **Comments:** | Each direction shall be tested separately | |

### 6.2.2 Test purposes for ISDN-PSTN

#### 6.2.2.1 Test purposes for ISDN-PSTN, basic call

| 410101 | Ref. to ETS 300 102-1 [1], subclause 5.1.6 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-PSTN/Basic_call/Successful/Speech/TC410101 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that the call establishment using en-bloc sending is performed correctly. During call establishment a Progress indicator information element shall be returned to the calling user with progress description value #1 "call is not end-to-end ISDN" or #2 "destination address is non-ISDN" | |
| **Parameter values:** | BC=speech, no HLC | |
| **Comments:** | P104 TC31101 | |

| 410102 | Ref. to ETS 300 102-1 [1], subclause 5.1.6 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-PSTN/Basic_call/Successful/Speech/TC410102 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that the call establishment using overlap sending is performed correctly. During call establishment a Progress indicator information element shall be returned to the calling user with progress description value #1 "call is not end-to-end ISDN" or #2 "destination address is non-ISDN" | |
| **Parameter values:** | BC=speech, no HLC | |
| **Comments:** | P104 TC31102 | |

| 410103 | Ref. to ETS 300 102-1 [1], subclause 5.3.3 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-PSTN/Basic_call/Successful/Speech/TC410103 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that the clearing procedure is performed correctly when the calling user clears after answer | |
| **Parameter values:** | BC=speech, no HLC | |
| **Comments:** | P104 TC31103 | |

| 410104 | Ref. to ETS 300 102-1 [1], subclause 5.3.3 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-PSTN/Basic_call/Successful/Speech/TC410104 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that the clearing procedure is performed correctly when the called user clears after answer | |
| **Parameter values:** | BC=speech, no HLC | |
| **Comments:** | P104 TC31104 | |

| 410105 | Ref. to ETS 300 102-1 [1], | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-PSTN/Basic_call/Successful/Speech/TC410105 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that the re-answer procedure is performed correctly when the called user clears and re-answers | |
| **Parameter values:** | BC=speech, no HLC | |
| **Comments:** | P104 TC31105 | |

| 410106 | Ref. to ETS 300 102-1 [1], subclauses 4.5.16, 5.1.6 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-PSTN/Basic_call/Successful/Speech/TC410106 | |
| **Selection criteria:** | | |
| **Test purpose:** | Support of telephony 3,1 kHz teleservice: Ensure that call establishment can be done with HLC. During call establishment a Progress indicator information element shall be returned to the calling user with progress description value #1 "call is not end-to-end ISDN" or #2 "destination address is non-ISDN" | |
| **Parameter values:** | BC=speech, HLC=telephony | |
| **Comments:** | P104 TC31106 | |

| 410301 | Ref. to ETS 300 102-1 [1], subclause 5.1.6 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-PSTN/Basic_call/Successful/Audio/TC410301 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that the call establishment using en-bloc sending is performed correctly. During call establishment a Progress indicator information element shall be returned to the calling user with progress description value #1 "call is not end-to-end ISDN" or #2 "destination address is non-ISDN" | |
| **Parameter values:** | BC=3,1 kHz audio, no HLC | |
| **Comments:** | P104 TC31201 | |

| 410302 | Ref. to ETS 300 102-1 [1], subclause 5.1.6 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-PSTN/Basic_call/Successful/Audio/TC410302 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that the call establishment using overlap sending is performed correctly. During call establishment a Progress indicator information element shall be returned to the calling user with progress description value #1 "call is not end-to-end ISDN" or #2 "destination address is non-ISDN" | |
| **Parameter values:** | BC=3,1 kHz audio, no HLC | |
| **Comments:** | P104 TC31202 | |

| 410303 | Ref. to ETS 300 102-1 [1], subclause 5.3.3 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-PSTN/Basic_call/Successful/Audio/TC410303 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that the clearing procedure is performed correctly when the calling user clears after answer | |
| **Parameter values:** | BC=3,1 kHz audio, no HLC | |
| **Comments:** | P104 TC31203 | |

| 410304 | Ref. to ETS 300 102-1 [1], subclause 5.3.3 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-PSTN/Basic_call/Successful/Audio/TC410304 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that the clearing procedure is performed correctly when the called user clears after answer | |
| **Parameter values:** | BC=3,1 kHz audio, no HLC | |
| **Comments:** | P104 TC31204 | |

| 410305 | Ref. to ETS 300 102-1 [1], subclauses 4.5.16, 5.1.6 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-PSTN/Basic_call/Successful/Audio/TC410305 | |
| **Selection criteria:** | | |
| **Test purpose:** | Support of Telefax G2/G3: Ensure that call establishment can be done with HLC. During call establishment a Progress indicator information element shall be returned to the calling user with progress description value #1 "call is not end-to-end ISDN" or #2 "destination address is non-ISDN" | |
| **Parameter values:** | BC=3,1 kHz audio, HLC=facsimile group 2/3 | |
| **Comments:** | P104 TC31205 | |

| 410306 | Ref. to ETS 300 102-1 [1], subclause 4.5.18 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-PSTN/Basic_call/Successful/Audio/TC410306 | |
| **Selection criteria:** | | |
| **Test purpose:** | Support of voice band data via modem: Ensure that call establishment can be done with LLC. During call establishment a Progress indicator information element shall be returned to the calling user with progress description value #1 "call is not end-to-end ISDN" or #2 "destination address is non-ISDN" | |
| **Parameter values:** | BC=3,1 kHz audio, LLC=voice band data via modem | |
| **Comments:** | P104 TC31206 | |

| 420101 | Ref. to ETS 300 102-1 [1] | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-PSTN/Basic_call/Unsuccessful/Speech/TC420101 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that when the called PSTN user is busy the network transport the cause value #17 "user busy" to the calling user | |
| **Parameter values:** | BC=speech | |
| **Comments:** | P104 TC32001 | |

| 420102 | Ref. to ETS 300 102-1 [1] | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-PSTN/Basic_call/Unsuccessful/Speech/TC420102 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that when calling to a unallocated PSTN number, the network initiate call clearing to the calling user with cause value #1 "unassigned number" | |
| **Parameter values:** | BC=speech | |
| **Comments:** | P104 TC32002 | |

| 420103 | Ref. to ETS 300 102-1 [1] | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-PSTN/Basic_call/Unsuccessful/Speech/TC420103 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that when the calling user clears with cause value #16 "normal call clearing" before answer from the called PSTN user, the call is cleared | |
| **Parameter values:** | BC=speech | |
| **Comments:** | P104 TC32003 | |

| 420104 | Ref. to ETS 300 102-1 [1] | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-PSTN/Basic_call/Unsuccessful/Speech/TC420104 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that when the called PSTN user is ringing but not answering, the network initiate call clearing to the calling user with cause value #18 "no user responding" or cause value #19 "no answer from user (user alerted)" | |
| **Parameter values:** | BC=speech | |
| **Comments:** | P104 TC32004 | |

| 420201 | Ref. to ETS 300 102-1 [1] | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-PSTN/Basic_call/Unsuccessful/UDI /TC420201 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that when the calling user requests digital connectivity for a call to a PSTN user, the network initiate call clearing to the calling user with cause value #88 "incompatible destination". | |
| **Parameter values:** | BC=UDI | |
| **Comments:** | | |

| 420301 | Ref. to ETS 300 102-1 [1] | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-PSTN/Basic_call/Unsuccessful/UDI/TC420301 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that when the called PSTN user is busy the network transport the cause value #17 "user busy" to the calling user | |
| **Parameter values:** | BC=3,1 kHz audio | |
| **Comments:** | | |

| 420302 | Ref. to ETS 300 102-1 [1] | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-PSTN/Basic_call/Unsuccessful/UDI/TC420302 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that when calling to a unallocated PSTN number, the network initiate call clearing to the calling user with cause value #1 "unassigned number" | |
| **Parameter values:** | BC=3,1 kHz audio | |
| **Comments:** | | |

| 420303 | Ref. to ETS 300 102-1 [1] | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-PSTN/Basic_call/Unsuccessful/UDI/TC420303 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that when the calling user clears with cause value #16 "normal call clearing" before answer from the called PSTN user, the call is cleared | |
| **Parameter values:** | BC=3,1 kHz audio | |
| **Comments:** | | |

| 420304 | Ref. to ETS 300 102-1 [1] | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-PSTN/Basic_call/Unsuccessful/UDI/TC420304 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that when the called PSTN user is ringing but not answering, the network initiate call clearing to the calling user with cause value #18 "no user responding" or cause value #19 "no answer from user (user alerted)" | |
| **Parameter values:** | BC=3,1 kHz audio | |
| **Comments:** | | |

#### 6.2.2.2 Test purposes for ISDN-PSTN, supplementary services

| 510001 | Ref. to ETS 300 286-1 [14], subclauses 11.2, 9.1.1.1.2 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-PSTN/Supplementary_services/Speech/UUS1/TC510001 | |
| **Selection criteria:** | The calling (served) user is provided with UUS1 implicit request | |
| **Test purpose:** | Ensure that when a User-user information element is included in the SETUP message sent from the calling user, call establishment can be done without User-user information. | |
| **Parameter values:** | BC=speech | |
| **Comments:** | P104 TC40001 | |

| 510002 | Ref. to ETS 300 097-1 [9], subclauses 9.5.1, 11 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-PSTN/Supplementary_services/Speech/COLP/TC510002 | |
| **Selection criteria:** | The calling user is provided with COLP | |
| **Test purpose:** | Ensure that the Connected number information element is network provided and correctly delivered to the calling user or, if the PSTN does not support this service, the presentation indicator indicate "number not available due to interworking" | |
| **Parameter values:** | BC=speech, SI=NP | |
| **Comments:** | P104 TC40002 | |

| 510003 | Ref. to ETS 300 098-1 [10], subclauses 9.3.1, 9.4.1, 11 | Other relevant ref.: ETS 300 097-1 [9], subclause 9.5.1 |
|---|---|---|
| **TSS reference:** | ISDN-PSTN/Supplementary_services/Speech/COLR/TC510003 | |
| **Selection criteria:** | The called PSTN user is provided with COLR, the calling user is provided with COLP | |
| **Test purpose:** | Ensure that the Connected number information element is network provided and delivered to the calling user without any digit information or, if the PSTN does not support this service, the presentation indicator indicate "number not available due to interworking" | |
| **Parameter values:** | BC=speech, (PI=PR), SI=NP, TON=unknown, NPI=unknown | |
| **Comments:** | P104 TC40003 | |

| 510005 | Ref. to ETS 300 138-1 [11] | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-PSTN/Supplementary_services/Speech/CUG/TC510005 | |
| **Selection criteria:** | The calling user belongs to a CUG with outgoing access allowed and called PSTN user is not member of the CUG | |
| **Test purpose:** | Ensure that when the calling user belongs to a CUG with outgoing access "allowed", call establishment to a PSTN user is possible | |
| **Parameter values:** | BC=speech, CUG default request | |
| **Comments:** | P104 TC40005 | |

| 510006 | Ref. to ETS 300 207-1 [16] | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-PSTN/Supplementary_services/Speech/CFU/TC510006 | |
| **Selection criteria:** | The user A and the user C are in the ISDN. The user B is in the PSTN and is provided with CFU. | |
| **Test purpose:** | Ensure that when user A calls user B, the call is forwarded to user C and user A is notified of call diversion. | |
| **Parameter values:** | BC=speech, CF active | |
| **Comments:** | P104 TC40006 | |

| 510007 | Ref. to ETS 300 210-1 [17] | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-PSTN/Supplementary_services/Speech/FPH/TC510007 | |
| **Selection criteria:** | Called user is Freephone subscriber in PSTN | |
| **Test purpose:** | Ensure that call establishment to a PSTN Freephone subscriber is possible, and that the calling user is not charged. During call establishment a Progress indicator information element shall be received by the calling user with progress description #1 "call is not end-to-end ISDN" or #2 "destination address is non-ISDN" | |
| **Parameter values:** | BC=speech | |
| **Comments:** | P104 TC40007 | |

| 510008 | Ref. to ETS 300 138-1 [11], subclauses 9.2.2.2, 11 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | ISDN-PSTN/Supplementary_services/Speech/CUG/TC510008 | |
| **Selection criteria:** | Calling user belongs to a CUG with outgoing access not allowed and called PSTN user is not member of the CUG | |
| **Test purpose:** | Ensure when the calling user belongs to a CUG with outgoing access "not allowed" and called PSTN user is not member of the CUG, call establishment is not possible and the network initiate call clearing to the calling user indicating cause value #87 "user not a member of CUG". | |
| **Parameter values:** | BC=speech, CUG default request | |
| **Comments:** | P104 TC32005 | |

| 510009 | Ref. to ETS 300 286-1 [14], subclauses 11.2, 9.1.1.2.2 | Other relevant ref.: ETS 300 102-1 [1], subclause 7.1.3.6 |
|---|---|---|
| **TSS reference:** | ISDN-PSTN/Supplementary_services/Speech/UUS1/TC510009 | |
| **Selection criteria:** | The calling (served) user is provided with UUS1 explicit request | |
| **Test purpose:** | Ensure that when the calling user explicit request UUS1 indicating "required", the network initiate call clearing to the calling user indicating cause value #69 "requested facility not implemented" or cause value #29 "facility rejected", and a UUS service 1 rejection with Error value "rejectedByUser". | |
| **Parameter values:** | BC=speech | |
| **Comments:** | P104 TC32006 | |

### 6.2.3 Test purposes for PSTN-ISDN

#### 6.2.3.1 Test purposes for PSTN-ISDN, basic call

| 610101 | Ref. to ETS 300 102-1 [1], subclauses 5.2.6, B.4 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | PSTN-ISDN/Basic_call/Successful/TC610101 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that call is delivered to the called ISDN user with the Bearer capability information element indicating "3,1 kHz audio". During call establishment a Progress indicator information element shall be included in the SETUP message sent to the called user with progress description value #1 "call is not end-to-end ISDN" or #3 "origination address is non-ISDN" | |
| **Parameter values:** | | |
| **Comments:** | P104 TC51001 | |

| 610102 | Ref. to ETS 300 102-1 [1] | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | PSTN-ISDN/Basic_call/Successful/TC610102 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that the clearing procedure is performed correctly when the calling user clears the call after answering. | |
| **Parameter values:** | | |
| **Comments:** | P104 TC51002 | |

| 610103 | Ref. to ETS 300 102-1 [1] | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | PSTN-ISDN/Basic_call/Successful/TC610103 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that the clearing procedure is performed correctly when the called ISDN user clears the call after answering | |
| **Parameter values:** | | |
| **Comments:** | P104 TC51003 | |

| 620101 | Ref. to ETS 300 102-1 [1] | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | PSTN-ISDN/Basic_call/Unsuccessful/TC620101 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that when the called ISDN user is busy, the calling user receives in-band information that the called user is busy | |
| **Parameter values:** | | |
| **Comments:** | P104 TC52001 | |

| 620102 | Ref. to ETS 300 102-1 [1] | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | PSTN-ISDN/Basic_call/Unsuccessful/TC620102 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that when the called ISDN user terminal is not connected, the calling user receives in-band announcement that the destination is out of order | |
| **Parameter values:** | | |
| **Comments:** | P104 TC52002 | |

| 620103 | Ref. to ETS 300 102-1 [1] | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | PSTN-ISDN/Basic_call/Unsuccessful/TC620103 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that when calling to unallocated ISDN number, the calling user receives in-band information that the called number is unallocated | |
| **Parameter values:** | | |
| **Comments:** | P104 TC52003 | |

| 620104 | Ref. to ETS 300 102-1 [1] | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | PSTN-ISDN/Basic_call/Unsuccessful/TC620104 | |
| **Selection criteria:** | Multipoint configuration for the called side. | |
| **Test purpose:** | Ensure that when the calling user clears before answer from the called ISDN user in a point-to-multipoint access configuration, the call is cleared | |
| **Parameter values:** | | |
| **Comments:** | P104 TC52004 | |

| 620106 | Ref. to ETS 300 102-1 [1] | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | PSTN-ISDN/Basic_call/Unsuccessful/TC620106 | |
| **Selection criteria:** | Point-to-point configuration for the called side. | |
| **Test purpose:** | Ensure that when the calling user clears before answer from the called ISDN user in a point-to-point access configuration, the call is cleared | |
| **Parameter values:** | | |
| **Comments:** | | |

| 620105 | Ref. to ETS 300 102-1 [1] | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | PSTN-ISDN/Basic_call/Unsuccessful/TC620105 | |
| **Selection criteria:** | | |
| **Test purpose:** | Ensure that when the called ISDN user is alerted by not answering before timer Q118 expires, the network initiate call clearing. | |
| **Parameter values:** | | |
| **Comments:** | P104 TC52005 | |

### 6.2.3.2 Test purposes for PSTN-ISDN, supplementary services

| 710001 | Ref. to ETS 300 092-1 [7], subclauses 9.5.1, 11 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | PSTN-ISDN/Supplementary_services/CLIP/TC710001 | |
| **Selection criteria:** | The called (served) user is provided with CLIP | |
| **Test purpose:** | Ensure that the Calling party number information element is network provided and correctly delivered to the called ISDN user or, if the PSTN does not support this service, the presentation indicator indicates "number not available due to interworking" | |
| **Parameter values:** | SI=NP | |
| **Comments:** | P104 TC53001 | |

| 710002 | Ref. to ETS 300 093-1 [8], subclause 9.4.1 | Other relevant ref.: ETS 300 092-1 [7], 9.5.1 |
|---|---|---|
| **TSS reference:** | PSTN-ISDN/Supplementary_services/CLIR/TC710002 | |
| **Selection criteria:** | The calling (served) user is provided with CLIR, the called user with CLIP | |
| **Test purpose:** | Ensure that the Calling party number information element is network provided and correctly delivered to the called user without any digit information or, if the PSTN does not support this service, the presentation indicator indicates "number not available due to interworking" | |
| **Parameter values:** | SI=NP, (PI=PR), TON=unknown, NPI=unknown | |
| **Comments:** | P104 TC53002 | |

| 710004 | Ref. to ETS 300 207-1 [16] | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | PSTN-ISDN/Supplementary_services/CFU/TC710004 | |
| **Selection criteria:** | The user A and the user C are in PSTN. The user B is in the ISDN and is provided with CFU. | |
| **Test purpose:** | Ensure that when user A calls user B, the call is forwarded to user C. | |
| **Parameter values:** | CF active | |
| **Comments:** | P104 TC53004 (or TC60004) | |

| 710005 | Ref. to ETS 300 130-1 [18] | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | PSTN-ISDN/Supplementary_services/MCID/TC710005 | |
| **Selection criteria:** | Called user is provided with MCID | |
| **Test purpose:** | Ensure that the call to an ISDN user is registered when the MCID service is requested by the called user in the Active call state | |
| **Parameter values:** | | |
| **Comments:** | P104 TC53005 | |

| 710006 | Ref. to ETS 300 210-1 [17] | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | PSTN-ISDN/Supplementary_services/FPH/TC710006 | |
| **Selection criteria:** | Called user is Freephone subscriber | |
| **Test purpose:** | Ensure that call establishment to an ISDN Freephone subscriber is possible, and that the calling user is not charged. During call establishment a Progress indicator information element shall be included in the SETUP message sent to the called user with progress description #1 "call is not end-to-end ISDN" or #3 "origination address is non-ISDN" | |
| **Parameter values:** | | |
| **Comments:** | P104 TC53006 | |

| 710007 | Ref. to ETS 300 138-1 [11], subclause 9.2.3 | Other relevant ref.: |
|---|---|---|
| **TSS reference:** | PSTN-ISDN/Supplementary_services/CUG/TC710007 | |
| **Selection criteria:** | Called user belongs to a CUG with incoming access not allowed and calling user is not member of the CUG | |
| **Test purpose:** | Ensure that when the called ISDN user belongs to a CUG with incoming access "not allowed" and calling user is not member of the CUG, the call is not established | |
| **Parameter values:** | | |
| **Comments:** | P104 TC52006 | |

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